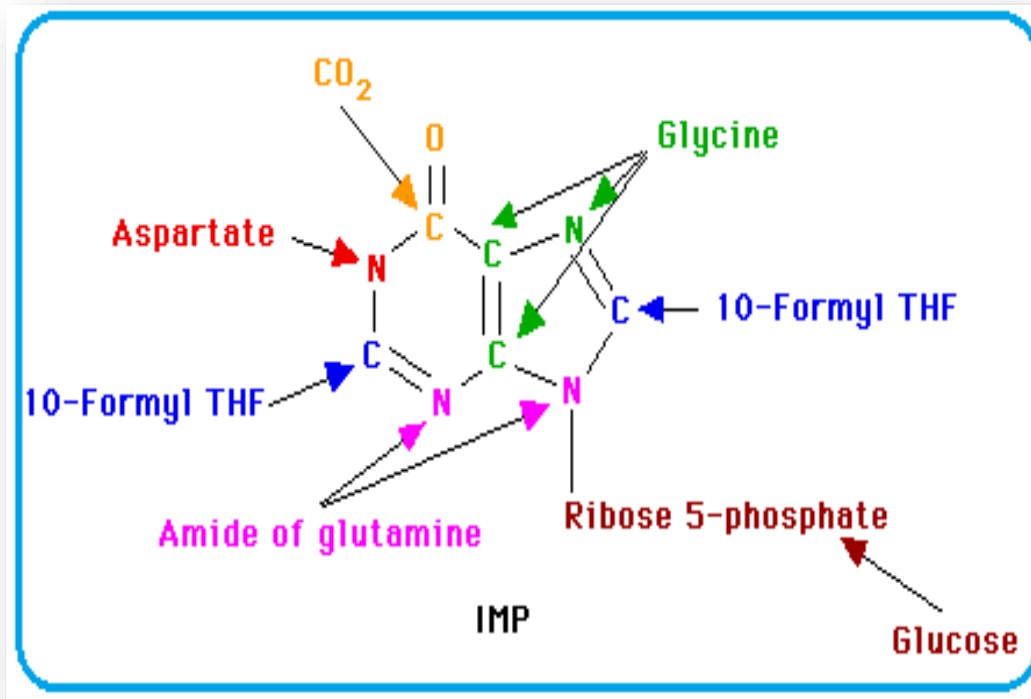


Purine Biosynthesis

- The purine bases are produced by *de novo* pathways:
 - ✓ uses **amino acids**, CO_2 , and N^{10} -formyl tetrahydrofolate as **precursors** to produce nucleotides
- Most *de novo* synthesis occurs:
 - ✓ in the **liver**, and
 - ✓ the **nitrogenous bases** and **nucleosides** are then transported to other tissues by **red blood cells**
- The **brain** also synthesizes **significant amounts of nucleotides**
- B/c **de novo pathway** requires:
 - ✓ **More than six high-energy bonds** per purine produced
- **A salvage pathway used by many cell types**,
 - ✓ can convert **free bases** and **nucleosides** to **nucleotides**

De novo Purine Nucleotide Biosynthesis

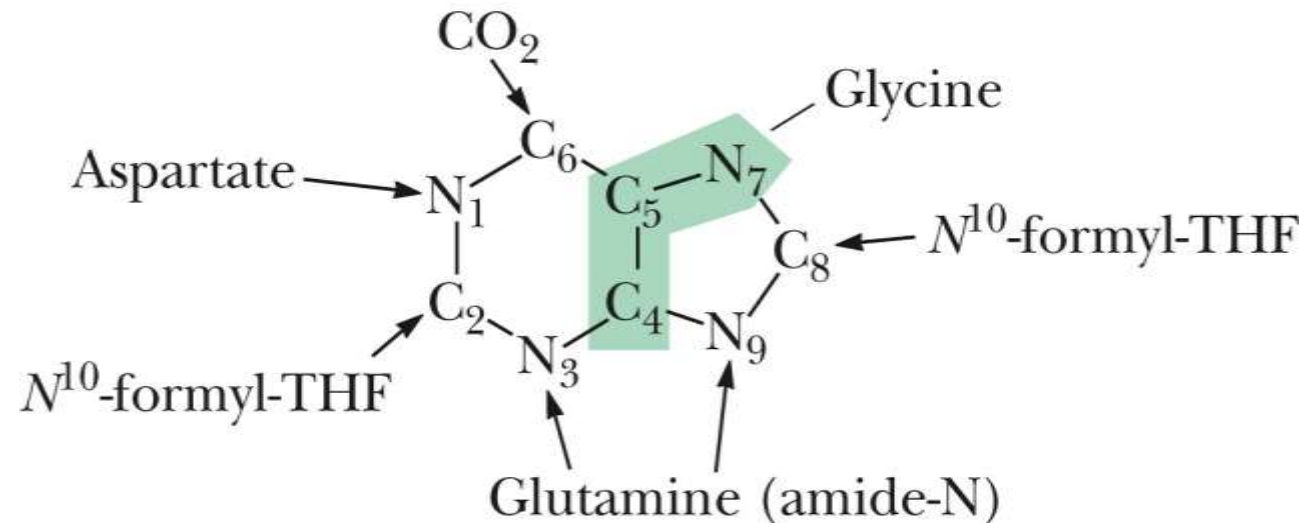
- The two parent purine nucleotides of nucleic acids are:
 - ✓ adenosine 5-monophosphate (AMP) and
 - ✓ guanosine 5-monophosphate (GMP).
- The **origin** of the **carbon and nitrogen atoms** of the **purine ring system**, as determined by **John Buchanan** using isotopic tracer experiments in birds



How Do Cells Synthesize Purines?

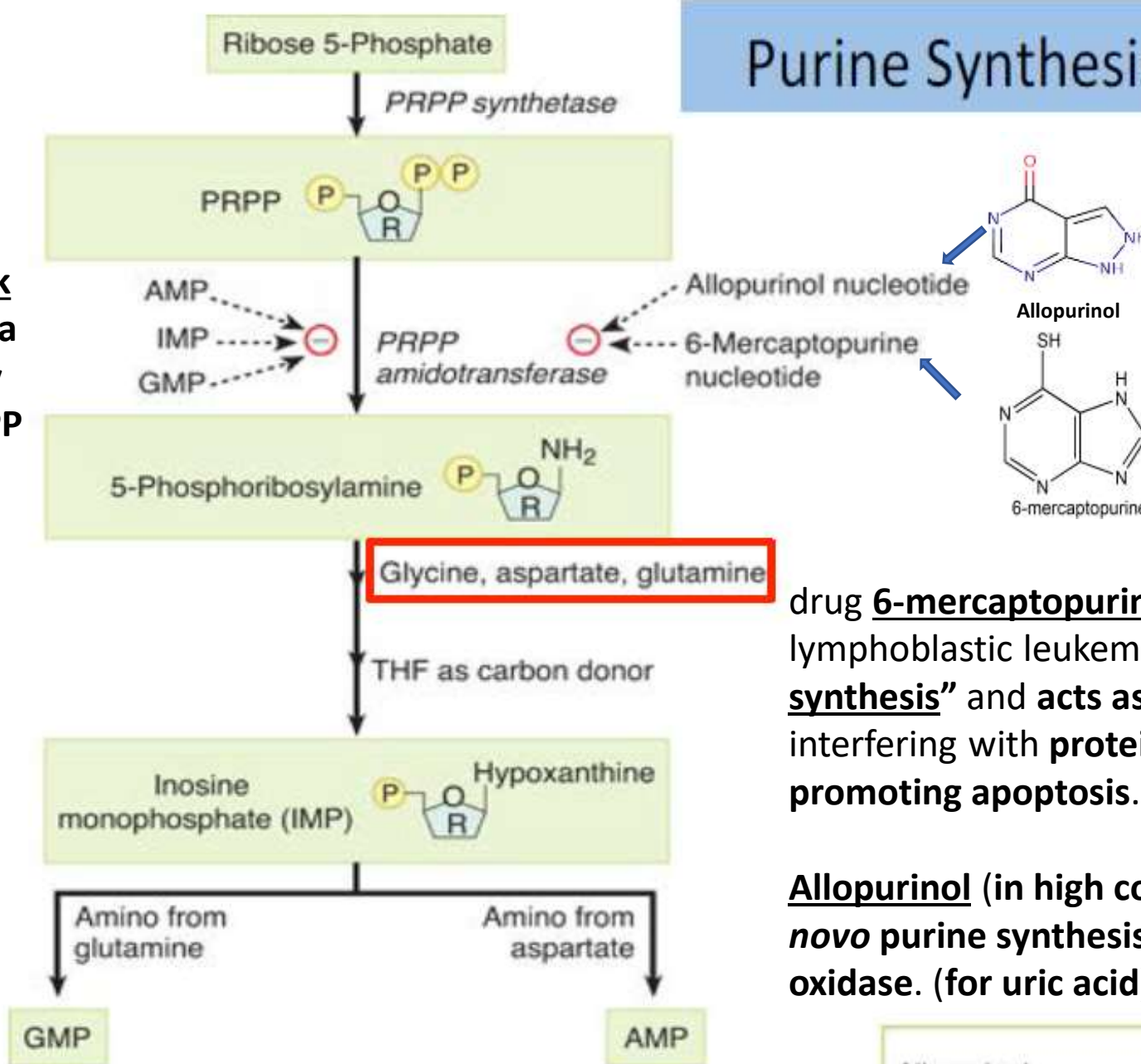
▪ *John Buchanan (1948) "traced" the sources of all nine atoms of the purine ring*

- N-1: aspartic acid
- N-3, N-9: glutamine (amide-N)
- C-2, C-8: N^{10} -formyl-THF - one carbon units
- C-4, C-5, N-7: glycine
- C-6: CO_2



Purine Synthesis Pathway

End product, feedback inhibition (binding to a different site) to a key anabolic enzyme (PRPP amidotransferase)



drug **6-mercaptopurine (6-MP)** → **Anticancer** (acute lymphoblastic leukemia) "**inhibits de novo purine synthesis**" and acts as an "**antiproliferative agent**" by interfering with **protein, DNA, and RNA synthesis** and **promoting apoptosis**.

Allopurinol (in high concentration) "**inhibits de novo purine synthesis**" in the **absence of xanthine oxidase**. (for uric acid)

HGPRT (recycle Ntds) → the two drugs also inhibit the **salvage** pathway →



- Step 1: Ribose-5-phosphate pyrophosphokinase.
- Step 2: Glutamine phosphoribosyl pyrophosphate amidotransferase.
- Step 3: Glycinamide ribonucleotide (GAR) synthetase.
- Step 4: GAR transformylase.
- Step 5: FGAM synthetase (FGAR amidotransferase).
- Step 6: FGAM cyclase (AIR synthetase).
- Step 7: AIR carboxylase.
- Step 8: SAICAR synthetase.
- Step 9: adenylosuccinase.
- Step 10: AICAR transformylase.
- Step 11: IMP synthase.

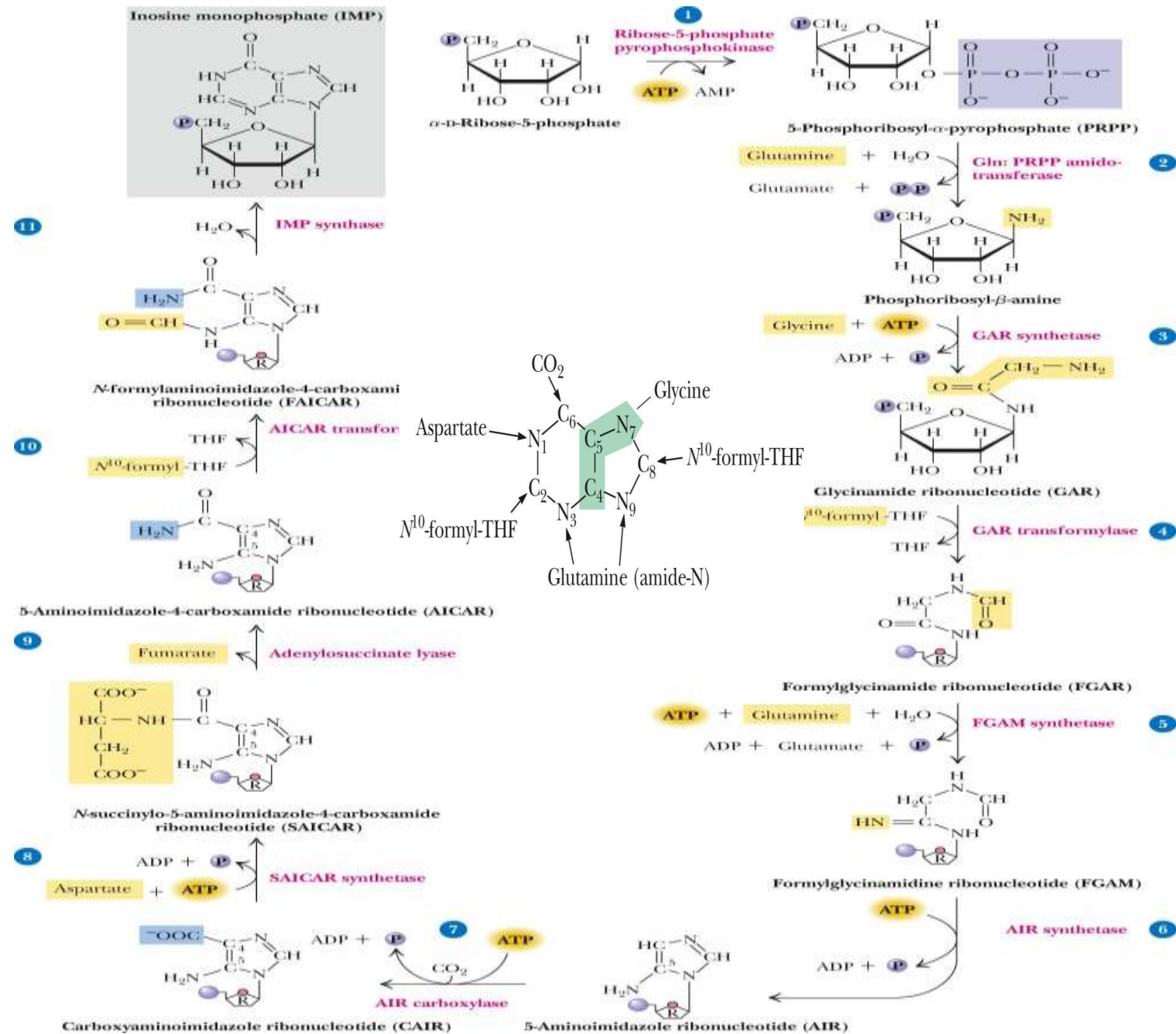
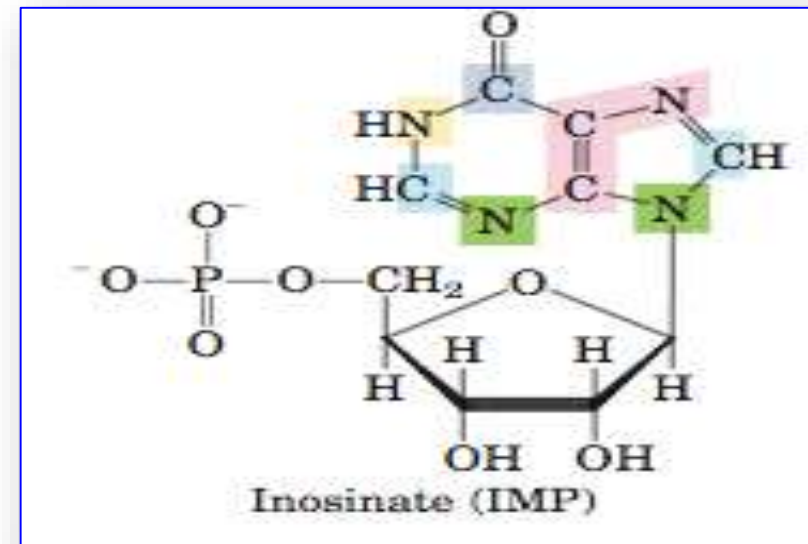


Figure
The *de novo* pathway for purine synthesis

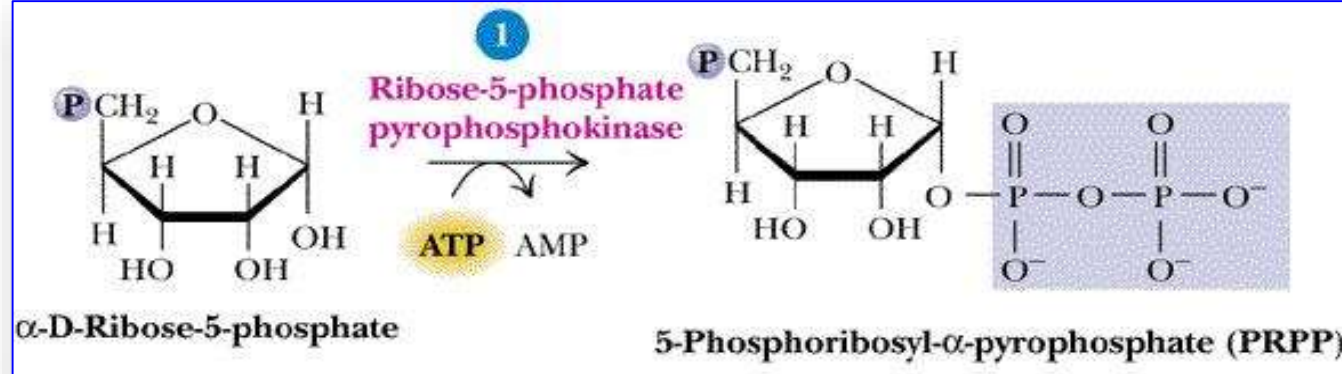
Synthesis of Inosine Monophosphate (IMP)

- Basic pathway for biosynthesis of **purine ribonucleotides**
- Starts from **ribose-5-phosphate** which is derived from the **pentose phosphate pathway**
- Requires **11 steps overall**
- Occurs primarily in the **liver**

IMP is the intracellular intermediate precursor for both (**AMP**) and (**GMP**) and thus plays a “central role in intracellular purine metabolism”



Step 1: Activation of ribose-5-phosphate



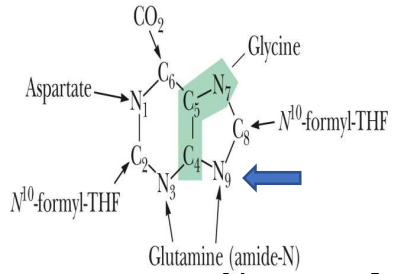
❖ R-5-P from PPP needs to be activated 1st by Ppi to **cyclic PRPP** (2 ATP requiring step) contains **2 phosphates**, by (RPPPK) enzyme, which requires Mg²⁺ and inorganic **phosphate (Pi)** as activators.

- **Enzyme:** Ribose-5-phosphate pyrophosphokinase
- **Product:** 5-phosphoribosyl- α -pyrophosphate (PRPP).
- This step is **tightly regulated by feedback inhibition**.
- PRPP is also a **precursor** in the biosynthesis of:
 - ✓ **pyrimidine** nucleotides and
 - ✓ the amino acids **histidine** & **tryptophan**.

The enzyme has an “**allosteric regulation site**”, activated by phosphate (substrate) and inhibited by ADP (low energy), both **compete** for the **allosteric site**

Step 2: Acquisition of purine atom 9

What is the properties of committed step



Second step: **Gln PRPP amidotransferase**

Form **phosphoribosyl- β -amine**;

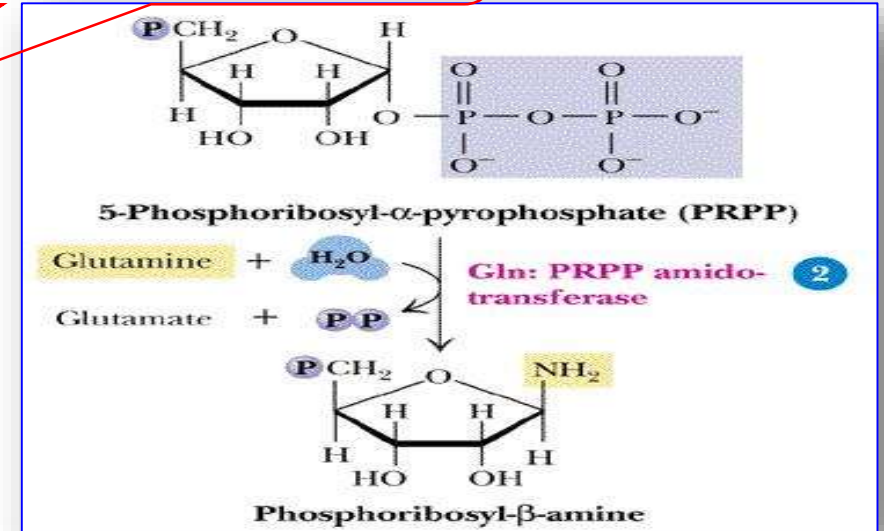
Changes C-1 configuration ($\alpha \rightarrow \beta$)

GMP and AMP inhibit this step - but at **distinct** sites

Azaserine - Gln analog -inhibitor/ **anti-tumor**

- Inhibit by **glutamine analogs**: azaserine and acivadin
- **Suicide inactivators**

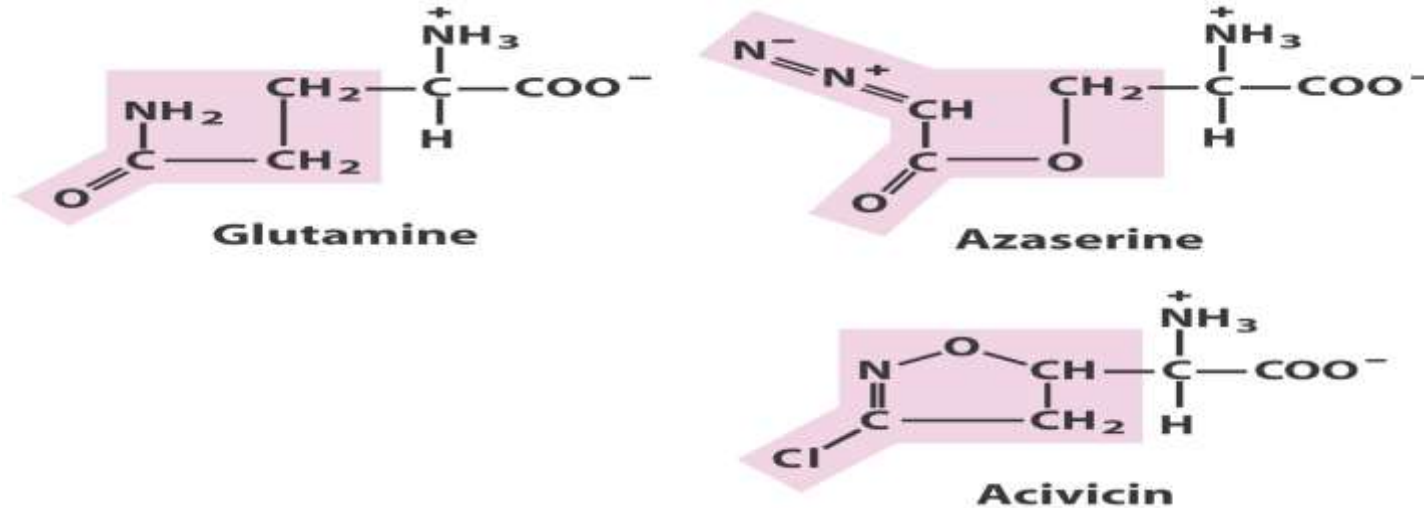
- In this **committed step**, an amino group donated by **glutamine** is attached at C-1 of PRPP.
- **Enzyme**: **Glutamine phosphoribosyl pyrophosphate amidotransferase**.
- **Product**: Resulting in the formation of **5-phosphoribosyl- β -amine**.



- PRPP is a branch point so this step is a committed step (highly regulated),
- Gln PRPP amidotransferase → is an allosteric enzyme regulated by AMP and GMP (allosteric regulation) feedback Inhibition
- PRPP itself → “feed-forward allosteric activation” to *Gln PRPP amidotransferase*

Cancer Chemotherapy:

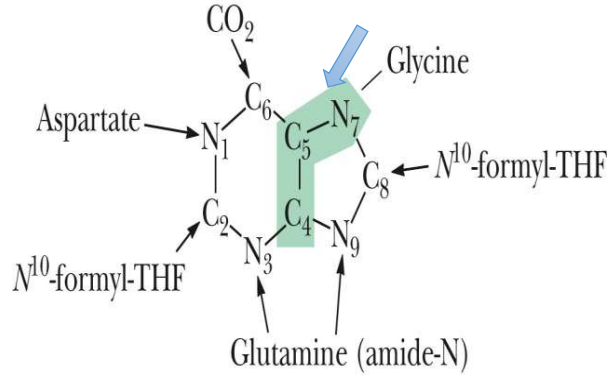
- Acts as an **irreversible inhibitor** of **glutamine-dependent enzymes**
 - ✓ by **covalently attaching** to nucleophilic groups in the **glutamine-binding site**.



→ **Azaserine**” anti-cancer, functions as a **purine antagonist** and **glutamine analog** that **competitively inhibits** pathways in which **glutamine is metabolized** (With **no formation** of product)

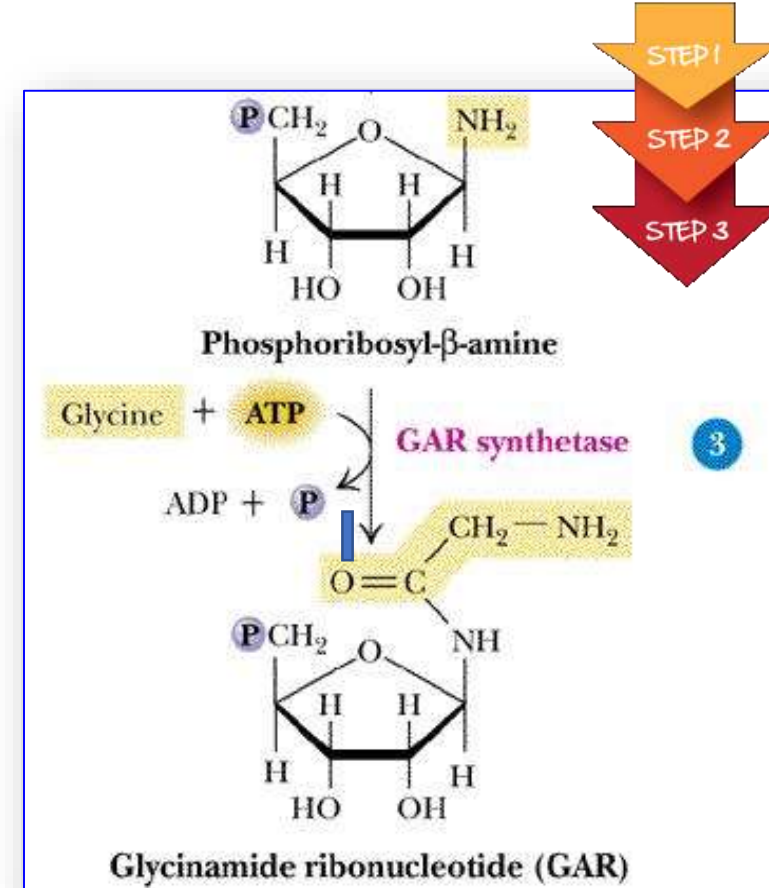
→ **Glutamine analog** → **Gln PRPP amidotransferase inhibitor**
“anti-cancer antibiotics” **irreversibly bind** to the **active site** .
→ Classical **cytotoxic immunosuppressants** act by inhibiting **DNA synthesis**. Suppress immune function.
→ Making the **enzyme site inaccessible** to **glutamine** (no addition of amino group) → leads to inhibition of the step.

Step 3: Acquisition of purine atoms C4, C5, and N7

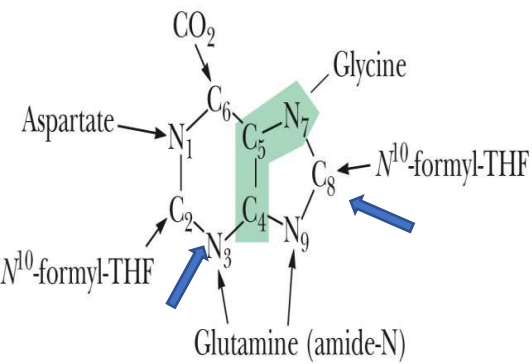


- All the 3 atoms are synthesis from a.a Glycine (clock-wise starting from atom 9)
- The “carboxy group of glycine” forms a bond with the “amino group” of substrate 5-phosphoribosyl-β-amine to form GAR

Synthetase “requires ATP” but not synthase (used in the synthesis of Bio mol without energy)



- Phosphoribosylamine reacts with ATP and glycine to produce **glycinamide ribonucleotide (GAR)**
- Enzyme: **GAR synthetase**



Step 4: Acquisition of purine atom C8

- Formylation of a free α -amino group of GAR.
- **Enzyme: GAR transformylase.**
- **Co-factor of enzyme: N¹⁰-formyl THF**

Step 5: Acquisition of purine atom N3

- The **amide amino group** of a second **glutamine** is transferred to form formylglycinamidine ribonucleotide (**FGAM**).
- The reaction is driven by the hydrolysis of **ATP**.
- **Enzyme: FGAM synthetase.**
- **Irreversibly inactivated** by **azaserine**

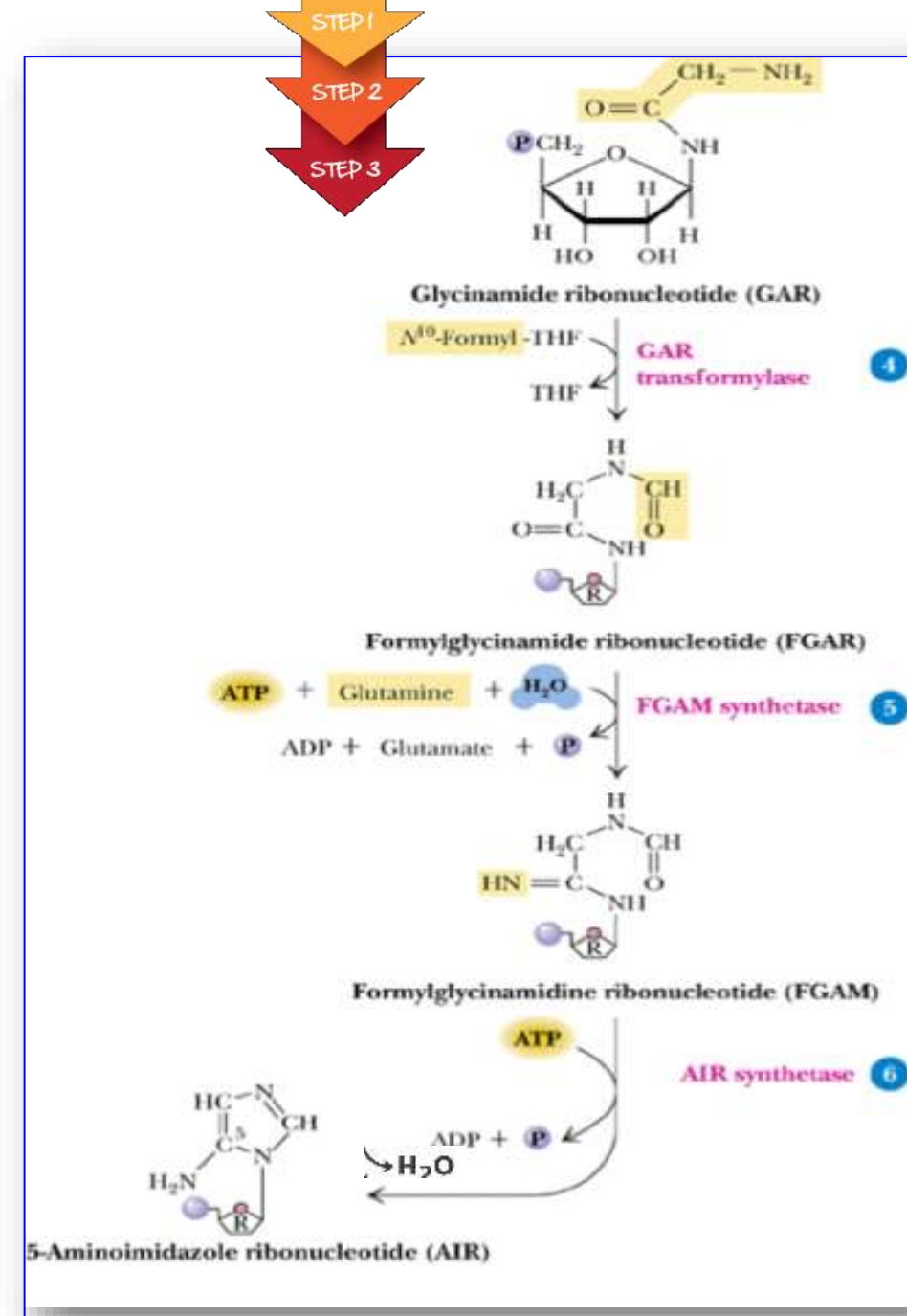
Step 6: Closing the ring

- Closing of the imidazole ring or formation of **5-aminoimidazole ribonucleotide (AIR)**.
- The reaction is driven by the hydrolysis of **ATP**.
- **Enzyme: AIR synthetase.**

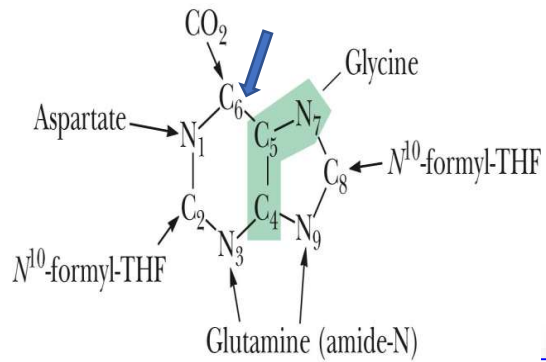
THF → source is **folic acid (Vit B9)** containing **diet**

F.A → DHF → THF (in the human body) by **THF reductase**, bacteria (Can synthesize it)

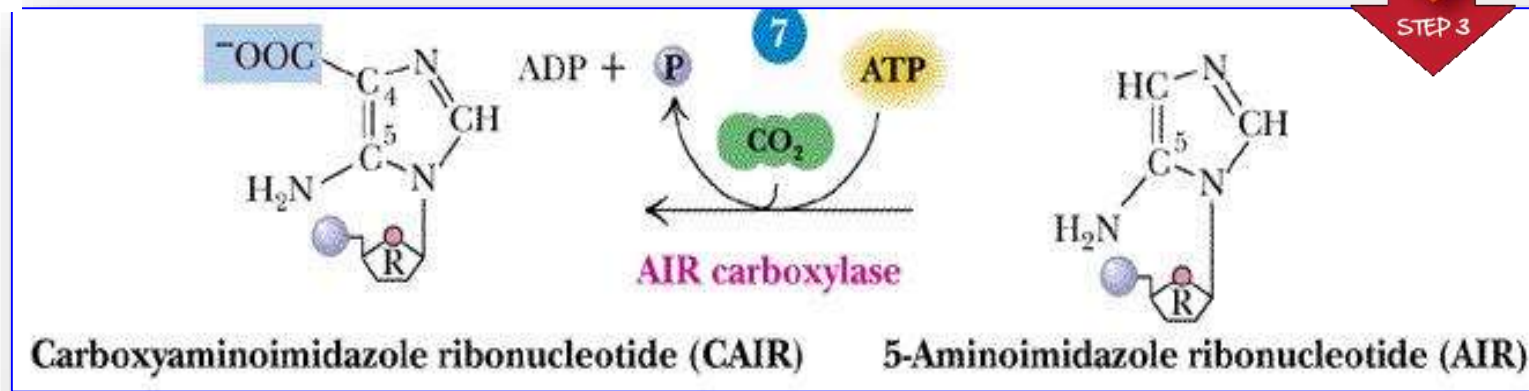
STEP 5 → “Azaserine (anti-cancer drug)” can Inhibit FGAM synthetase (step 5) similar to step 2, B/c its glutamine analog



- Bacteria (Can synthesize it, from **Pteridine** and PABA)
- In humans → THF source is folate (Vit B9) containing **diet vegetables** and **fortified food (folic acid is the synthetic form)**
- F.A → DHF → THF (in the human body) both steps by DHF reductase, require **NADPH**,
- Anti folate drugs methotrexate(MTX) (a potent anti-tumor drug that binds DHFR), N5-N10 Methylene THF for **thymine** synthesis from dUMP... antibiotic trimethoprim inhibit TMP formation



Step 7: Acquisition of C6



- C6 is introduced as HCO_3^-
- The reaction is driven by the hydrolysis of **ATP**
- **Enzyme:** **AIR carboxylase** (aminoimidazole ribonucleotide carboxylase)
- **Product:** **CAIR** (carboxyaminoimidazole ribonucleotide)

Step 8: Acquisition of N1

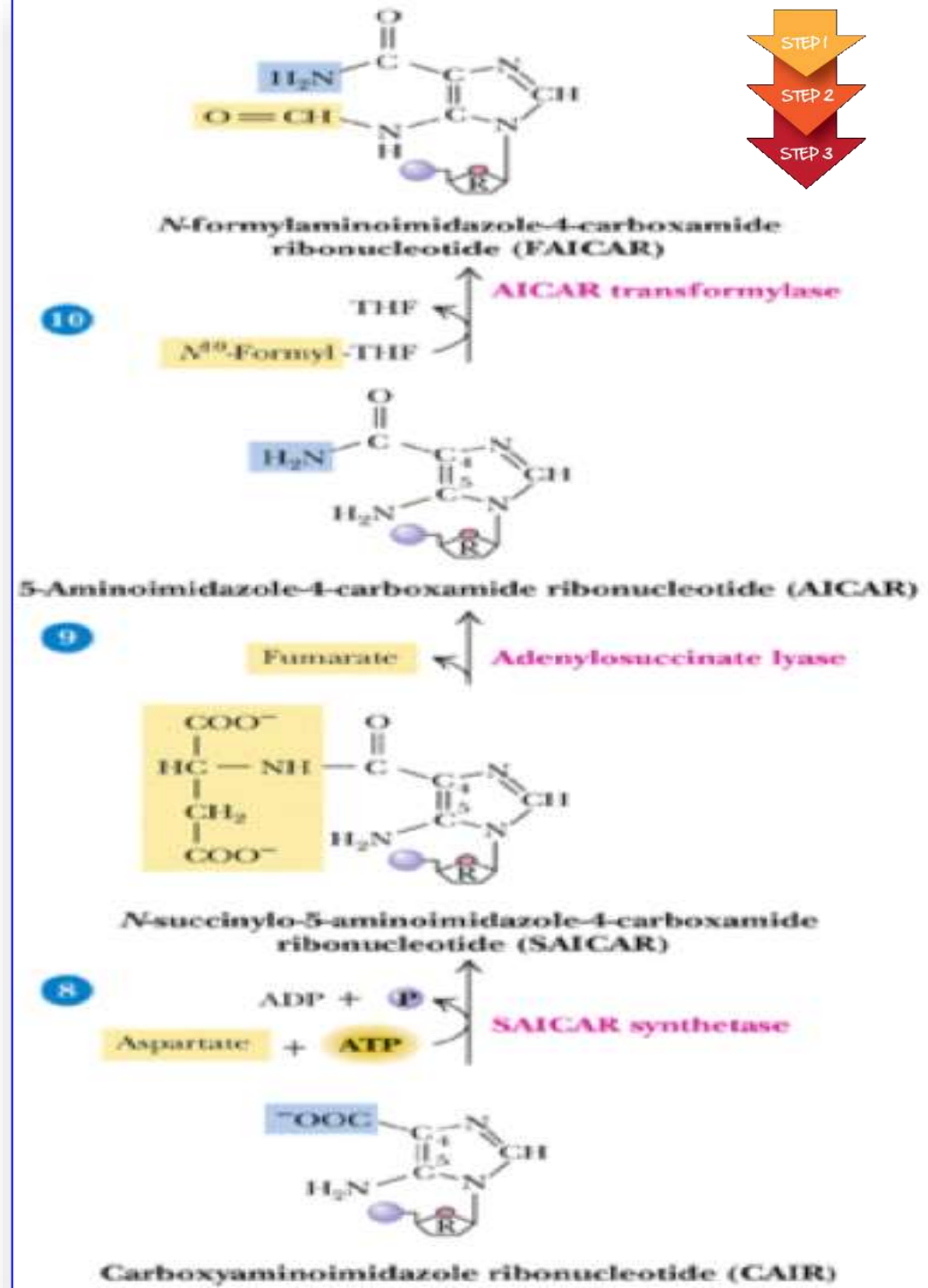
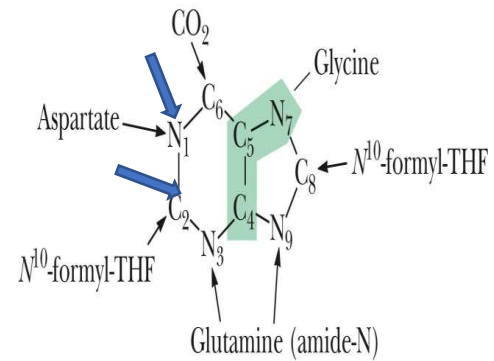
- N1 is acquired from **aspartate** in an amide condensation reaction that is driven by hydrolysis of **ATP** to produce N-succinyl-5-aminoimidazole-4-carboxamide ribonucleotide (SAICAR).
- Enzyme: SAICAR synthetase**

Step 9:

- Elimination of fumarate** by the action of **adenylosuccinate lyase** to produce 5-aminoimidazole-4-carboxamide ribonucleotide (AICAR)

Step 10: Acquisition of C2

- Another formylation reaction catalyzed by **AICAR transformylase** results in the formation of 5-formylaminoimidazole-4-carboxamide ribonucleotide (FAICAR)
- incorporation of a **formyl group** from **N10 formyl THF**
- Anti-cancer and anti-bacterial drug** work by inhibiting THF formation (inhibit this step)



Biochemistry in medicine



- The dependence of **purine biosynthesis** on **folic acid** compounds at Steps 4 and 10 means that:
 - ✓ **antagonists of folic acid metabolism indirectly inhibit purine formation** and,
 - ✓ in turn, ***inhibit nucleic acid synthesis, cell growth, and cell division***
- Clearly, **rapidly dividing cells** such as
 - ✓ **Malignancies (fast-growing tumors) or infective bacteria** are **more susceptible to these antagonists** than slower-growing normal cells
- The dependence of purine biosynthesis on **THF** in two steps means that **methotrexate** and **sulfonamides block purine synthesis**

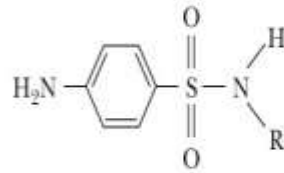
- **Folic acid (THF is active form)** → is **vitamin B9 water soluble**, THF is used for the **maturation of RBC** folate **deficiency** causes **megaloblastic anemia**, the source is **green vegetables, Lentin, oranges, cereal, beans, and the liver..** body needs it in a small amount to function
- **Cancer cell and bacteria** has a **faster rate of purine biosynthesis** than normal cell
- **“THF analog”** or **metabolism antagonist drug** is used for the treatment

Folate Analogs as Antimicrobial and Anticancer Agents

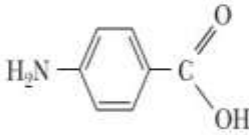
- *De novo* purine biosynthesis depends on **folic acid compounds** at steps 4 and 10
- **Sulfonamides** are effective anti-bacterial agents
- **Methotrexate and related compounds** **inhibit** the **reduction of dihydrofolate to tetrahydrofolate**, catalyzed by **dihydrofolate reductase**.
- **Methotrexate** and **Aminopterin** are folic acid analogs that have been used in **cancer chemotherapy** but are also toxic to all dividing cells
- **Methotrexate chemotherapy** is a **Competitive inhibitor of DHFR (folic acid analog) folate antagonist** → and therefore interferes with DNA synthesis, repair, and cellular replication. **Inhibit the 2 steps.**

(a)

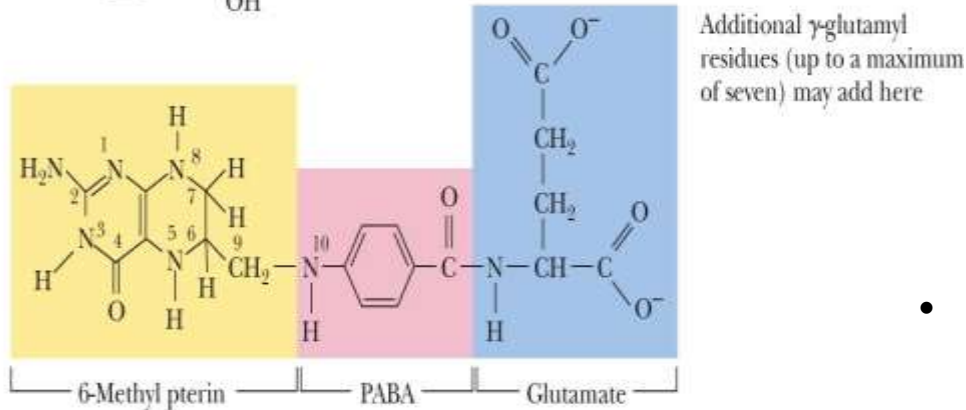
Sulfonamides have the generic structure:



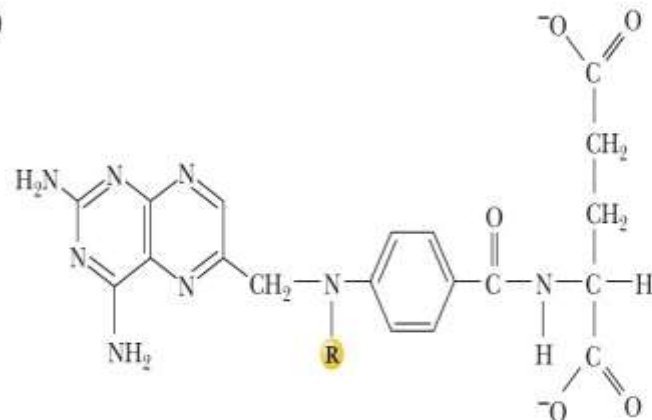
PABA (*p*-aminobenzoic acid)



THF (tetrahydrofolate)



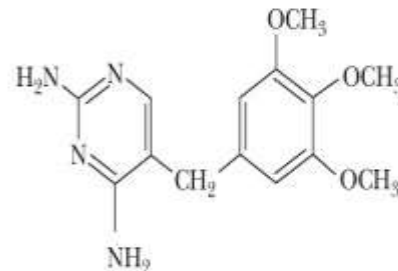
(b)



2-Amino, 4-amino analogs of folic acid

$\text{R} = \text{H}$ Aminopterin

$\text{R} = \text{CH}_3$ Amethopterin (methotrexate)

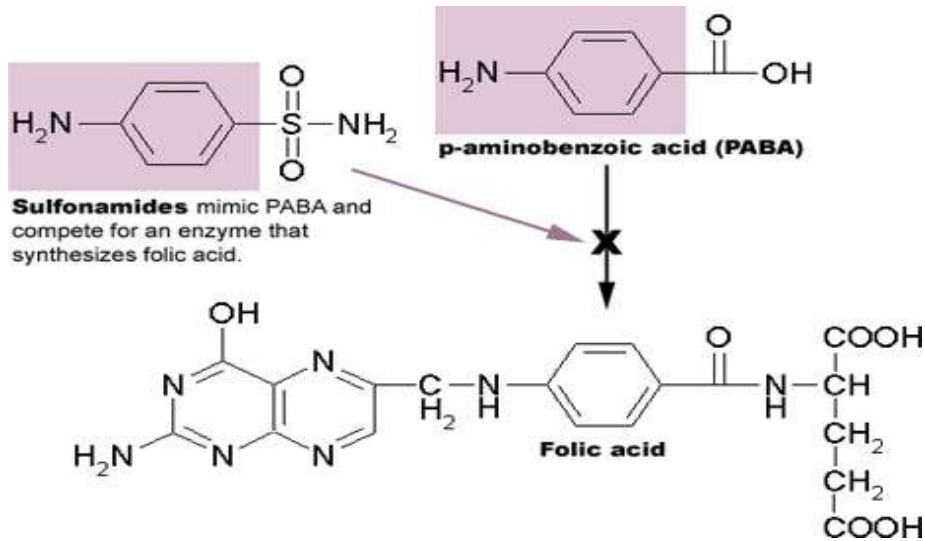


Trimethoprim

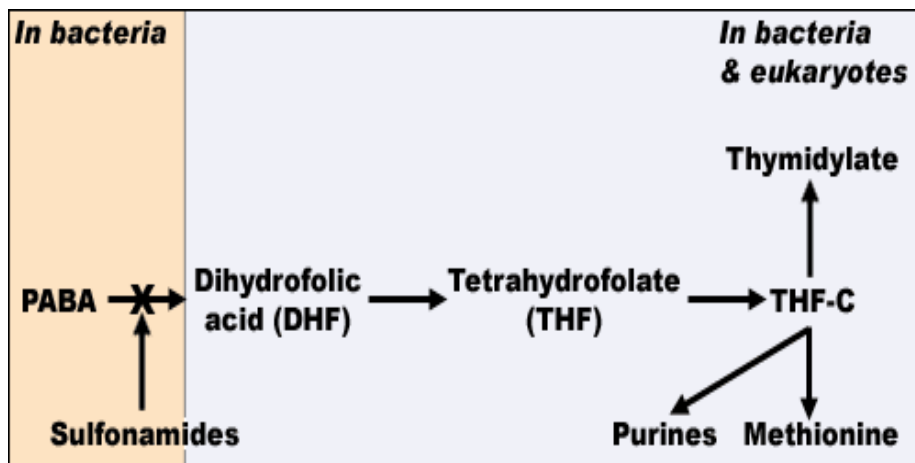
Sulfa drugs, or sulfonamides

- their antibiotic properties are due to **their similar to *p*-aminobenzoate (PABA)**, an important precursor in the folic acid synthesis
- Sulfonamides block folic acid formation by competing with PABA**
- Sulfa drugs (PABA analog) are classified as **anti-metabolite antibiotics**, inhibit dihydropteroate synthase
- Trimethoprim (DHF analog) is an anti-bacterial Inhibit the DHFR enzyme.
- Anti-cancer chemotherapy (antimetabolite) → binds to dihydrofolate reductase with about **1000-fold greater affinity** than **DHF** and thus **acts as a virtually irreversible inhibitor**.

- **p-aminobenzoic acid (PABA)** is required by many bacteria to produce an important enzyme cofactor, **THF**
- These bacteria require THF for their growth and division



- **Sulfanilamide** acts as a **competitive inhibitor** to **dihydropteroate synthase** that converts PABA into folic acid, resulting in a depletion of this cofactor.
- The depletion of this cofactor results in **retarded growth** and eventual **death of the bacteria**.

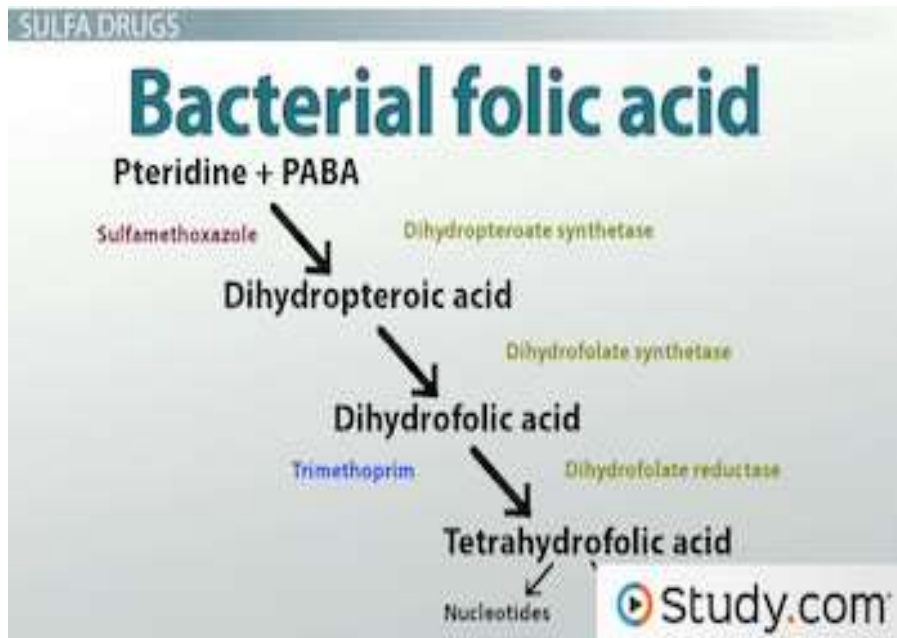


In Bacteria → A pteridine is an aromatic chemical condensed with PABA it forms DHF

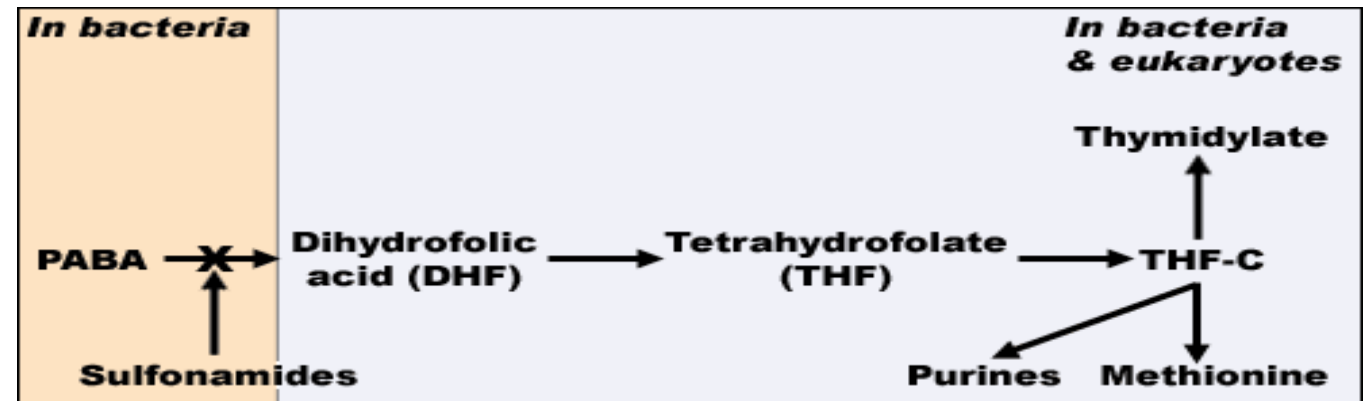
- Pteridine + PABA $\xrightarrow{\text{dihydropteroate synthase}}$ Dihydropteroic acid(DHP) $\xrightarrow{\text{dihydrofolate synthetase}}$ DHF $\xrightarrow{\text{DHFR}}$ THF

- THF → used in both purine and pyrimidine metabolism

- Mammals absorb their folic acid from their diets, so **sulfanilamide** exerts **no effects** on them.
- **Sulfonamide** selectively kills bacteria and is effective “without affecting human cells” because the **folic acid source is food**



- **Vitamin B9 cannot enter bacterial cells** and thus bacteria must make their own from **PABA** and **pteridine**



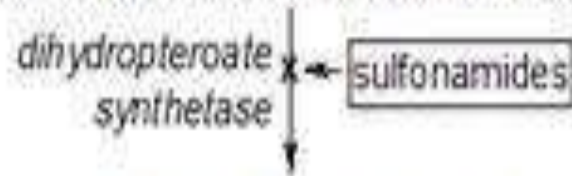
- **DHF reductase** catalyzes **both step** reactions in **mammals** forming **THF** active cofactor of enzymes

Bacteria: PABA → DHF → THF

Mammal: Folic acid (diet) → DHF → THF

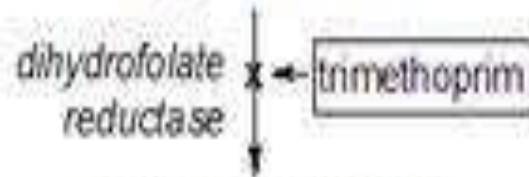
Sulfanilamide Compete for **dihydropteroate synthase**

dihydropteroate diphosphate + p-aminobenzoic acid (PABA)



dihydropteroic acid

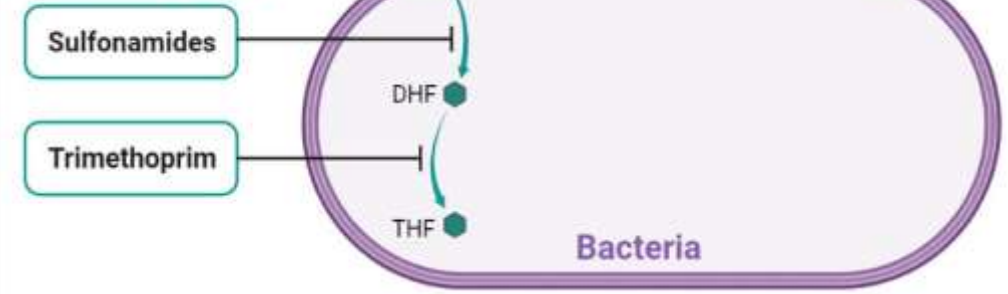
dihydrofolic acid



tetrahydrofolic acid

Folic acid synthesis inhibitors

Folic acid synthesis



para-aminobenzoic acid (PABA)

$\xrightarrow[\text{Sulfonamide competes with PABA for dihydropteroate synthetase}]{\text{dihydropteroate synthetase}}$

dihydrofolate

$\xrightarrow[\text{Trimethoprim blocks successive steps in the synthesis of reduced folic acid.}]{\text{dihydrofolate reductase}}$

tetrahydrofolate

Synthesis of thymidylate

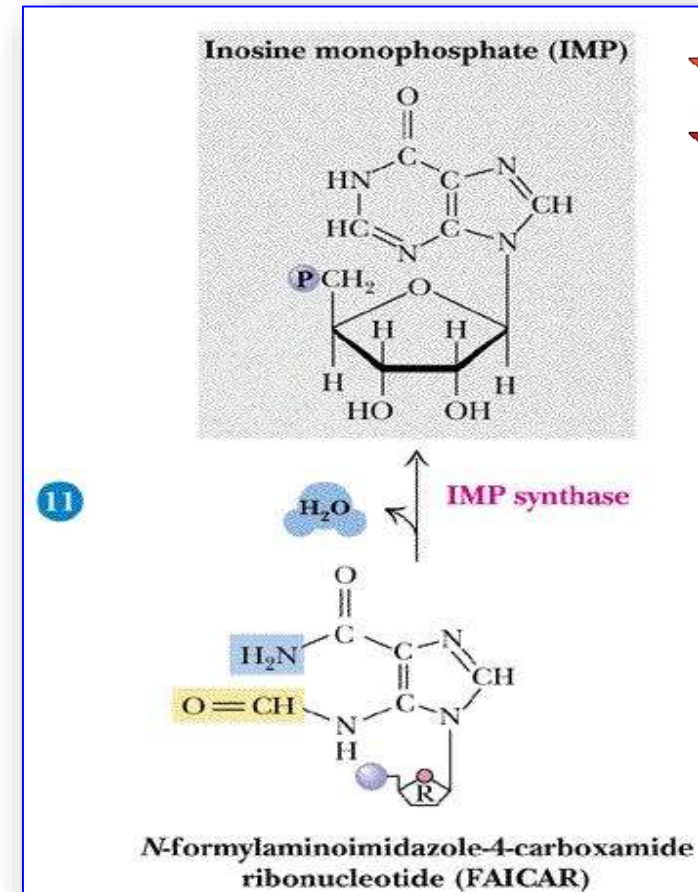
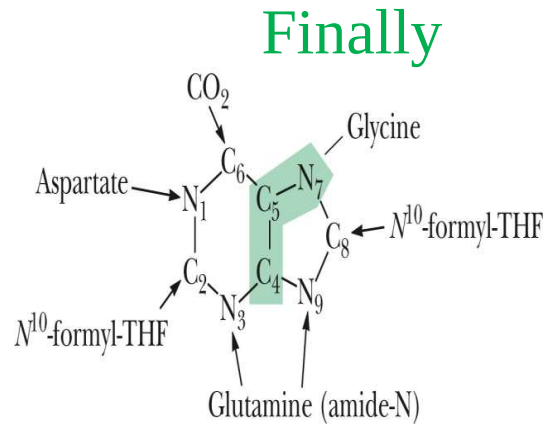
DNA

- Difference between trimethoprim and sulfa antibiotic drugs?
Sulfa → pteridine and PABA do not condense and form DHF due to **dihydropteroate synthase** (DHPS) inhibition
Trimethoprim → inhibits **DHFR** for **UTI**, occasionally for a chest infection, can be given along with sulfa, and has a synergetic effect (pneumonia in HIV patient)

...cont'd

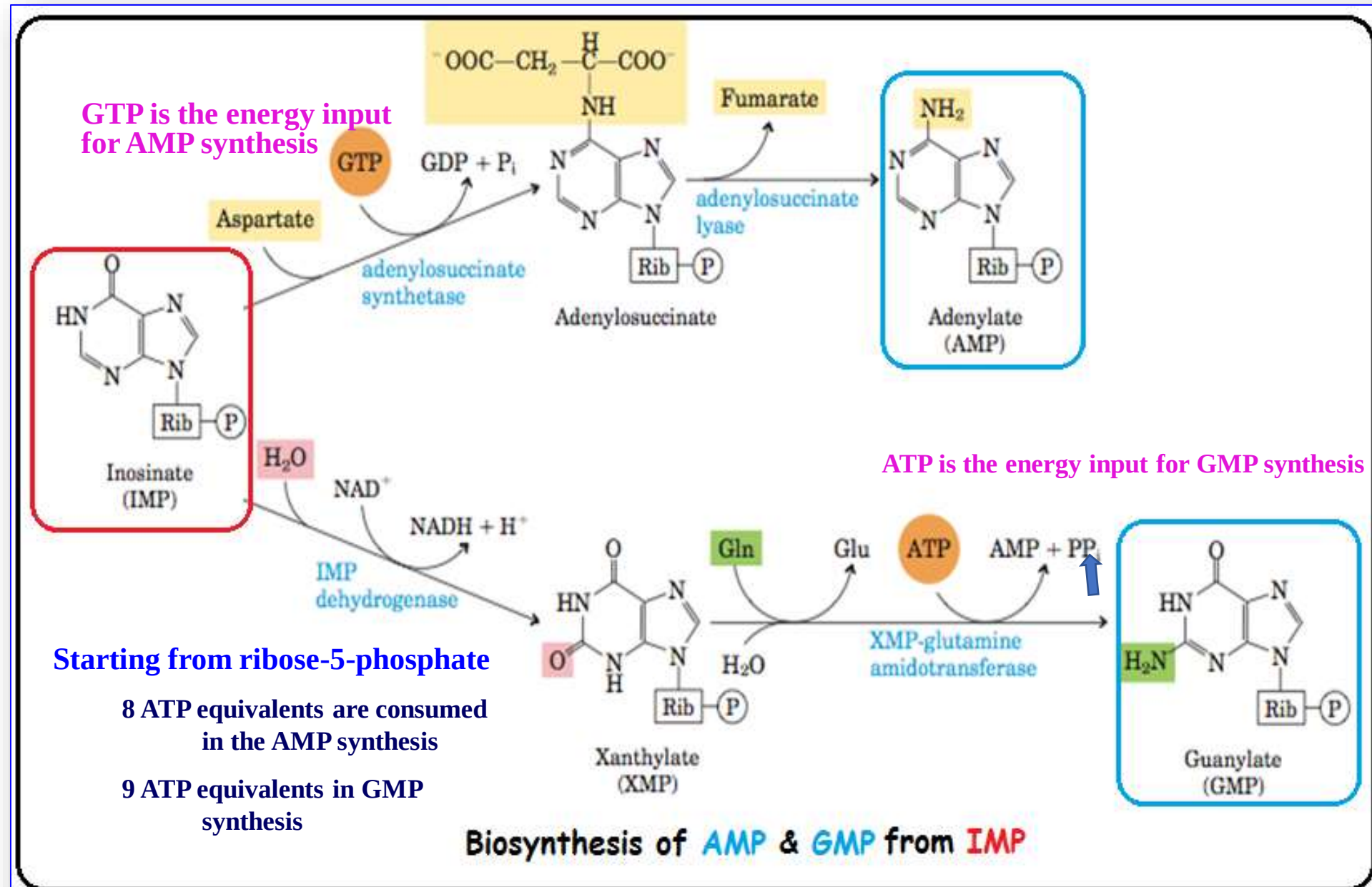
- Inhibitors of human purine synthesis are **extremely toxic to tissues**,
 - especially to developing structures such as in a fetus, or
 - to cell types that normally replicate rapidly,
 - Most of the chemo **side effects**... has **toxicity to normal dividing cell**
 - including those of bone marrow, skin, gastrointestinal (GI) tract, immune system, or hair follicles.
- As a result, individuals taking such **anticancer drugs can experience adverse effects, including**
 - **anemia,**
 - **scaly skin,**
 - **GI tract disturbance,**
 - **Immuno-deficiencies, and**
 - **hair loss.**

Step 11: Cyclization or ring closure

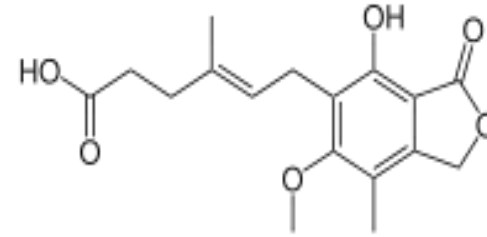


- **Water** is eliminated by the action of **inosine monophosphate (IMP) synthase**
- In contrast to step 6 (closure of the imidazole ring), this reaction **does not require ATP hydrolysis**
- Once IMP is formed, it is **rapidly converted to AMP and GMP**

Biosynthesis of AMP and GMP from IMP



Mycophenolic Acid



- **Inhibits** *inosine monophosphate dehydrogenase*
- Enzyme converts **IMP** to **xanthosine monophosphate** (to be converted to **GMP**)
- **B and T lymphocytes** need nucleotides to divide
- **Drug** is used to **prevent organ transplant graft rejection**
- **Mycophenolic Acid** is an antineoplastic antibiotic (immunosuppressant medication) derived from various **Penicillium** fungal species. Used during transplant, and prevent autoimmune (lupus)
- MPA, inhibits (IMPDH), preventing the formation of **GMP** and “synthesis of lymphocyte DNA” which results in the inhibition of lymphocyte proliferation, antibody production, cellular adhesion, and migration of T and B lymphocytes. To prevent rejection
- in patients receiving allogeneic renal transplants, administered in combination with cyclosporine and corticosteroids. To prevent graft rejection
- MPA also has antibacterial, antifungal, and antiviral activities. is an immunosuppressant drug and **potent anti-proliferative**,

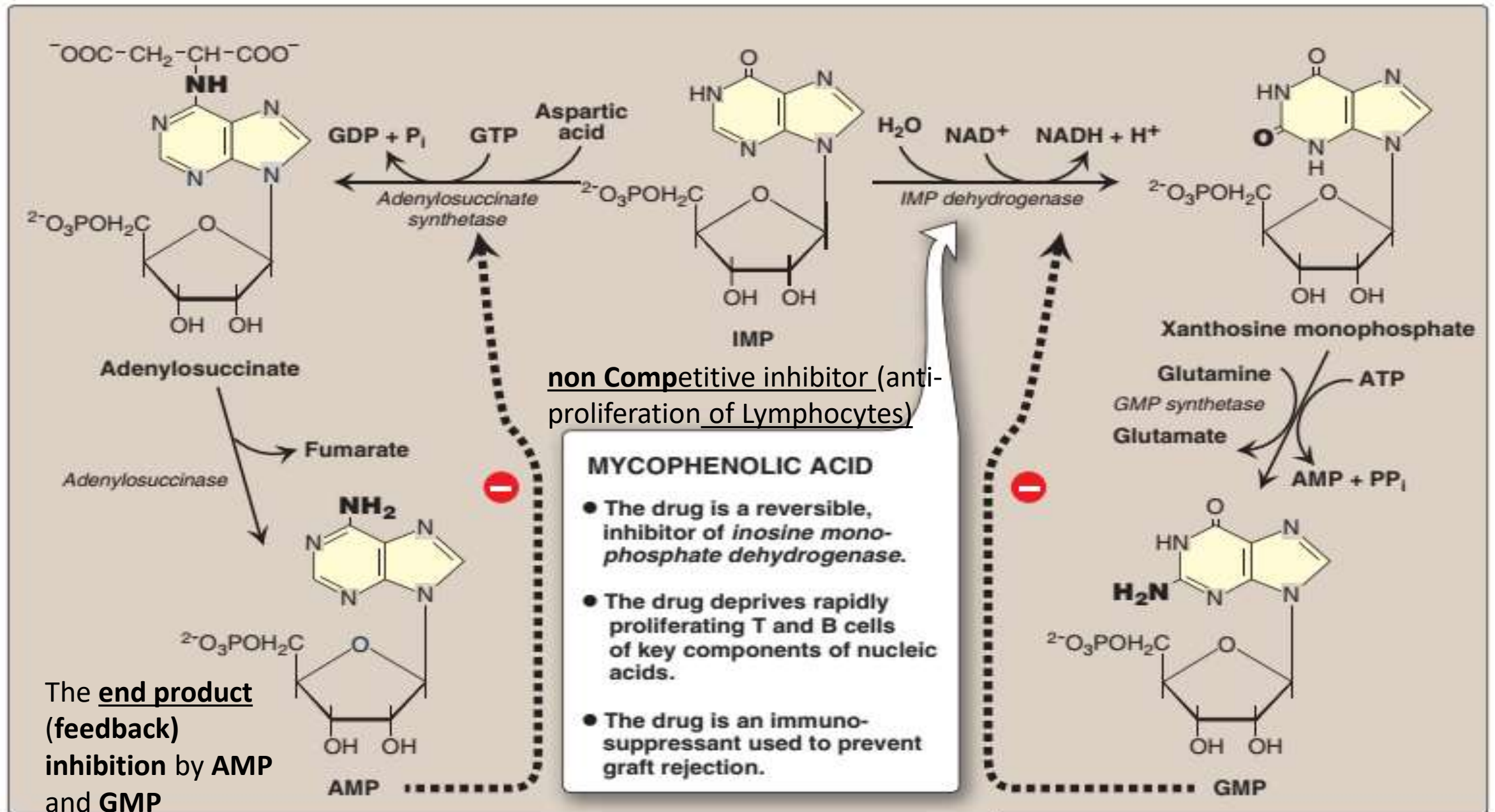


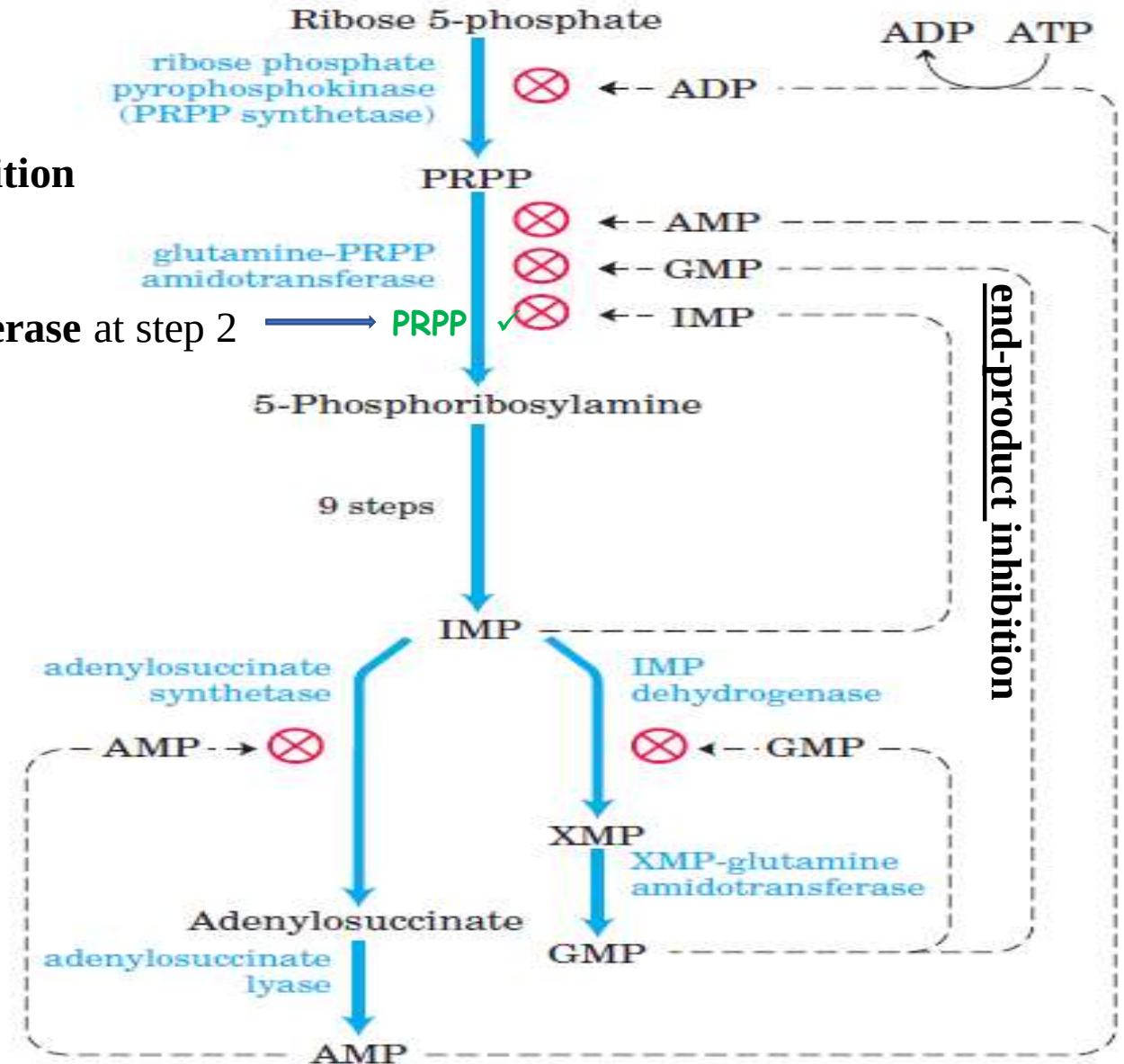
Fig. Conversion of IMP to AMP and GMP showing feedback inhibition

Purine nucleotide biosynthesis is regulated by feedback control

Purine synthesis **regulations** include....

- Mainly by feedback control → end-product inhibition
- High AMP allosterically inhibits the enzymes

PRPP is a feed-forward activator of ...Amido transferase at step 2 → PRPP ✓



The significance of regulation:

(1) Fulfill the need of the body, without wasting.

(2) $[GTP] = [ATP]$

Nucleoside diphosphate and triphosphate

Nucleoside diphosphate: **ATP-dependent kinase**

- *Adenylate kinase*: $\text{AMP} + \text{ATP} \rightarrow \text{ADP} + \text{ADP}$
- *Guanylate kinase*: $\text{GMP} + \text{ATP} \rightarrow \text{GDP} + \text{ADP}$

Nucleotide triphosphate: **non-specific enzyme**

it acts on “all types of NTds Di phosphate” NDP to NTP

- *Nucleoside diphosphate kinase*



The structure of nucleotides

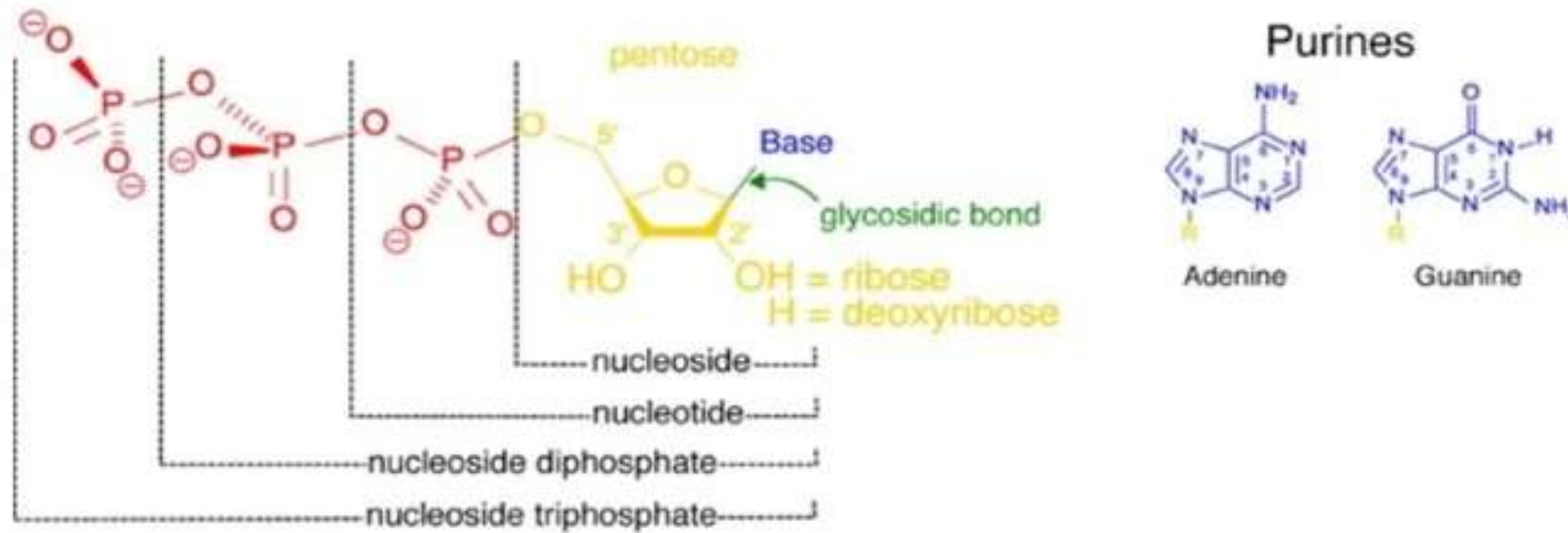


Figure 1. The structure of nucleotides depicts how the **base** and **pentose** sugar (nucleoside, in yellow, and bond in green) may be attached to either **1, 2, or 3 phosphate groups** (red)

A nucleoside attached to **1 phosphate** (red) is a **nucleoside monophosphate**

The addition of **2nd phosphate** group (red) forms a **nucleoside diphosphate**, and finally, the addition of a **3rd phosphate** forms a **nucleoside triphosphate**

How Do Cells Form the Deoxyribonucleotides That Are Necessary for DNA Synthesis?

- Reduction at **2'**-position of ribose ring
- Serve as a precursor for DNA synthesis
- Replacement of 2'-OH with *hydride* is catalyzed by **ribonucleotide reductase**
 - An $\alpha_2\beta_2$ -type enzyme - subunits R_1 (86 kD) and R_2 (43.5 kD)
 - R_1 has two regulatory sites, a **specificity site**, and an overall **activity site**

“ribonucleotide reductase” → reduction at C2 (become deoxy)...

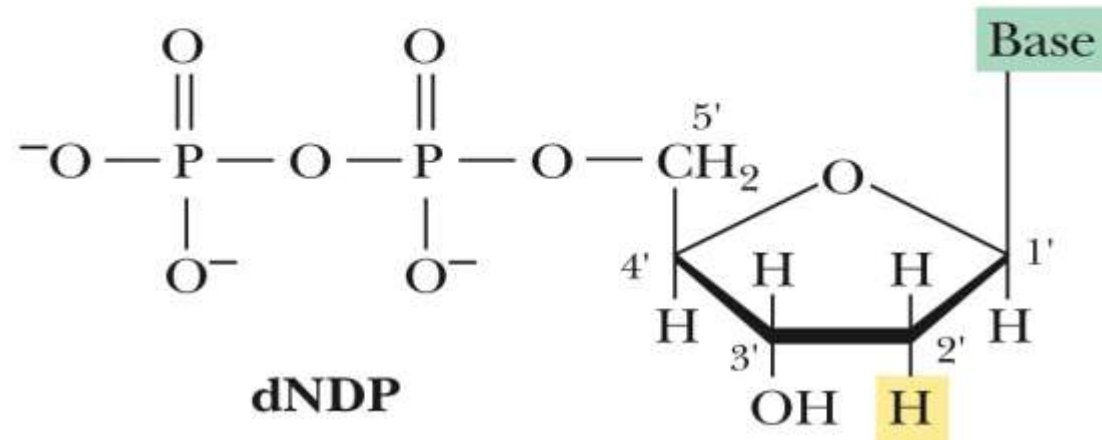
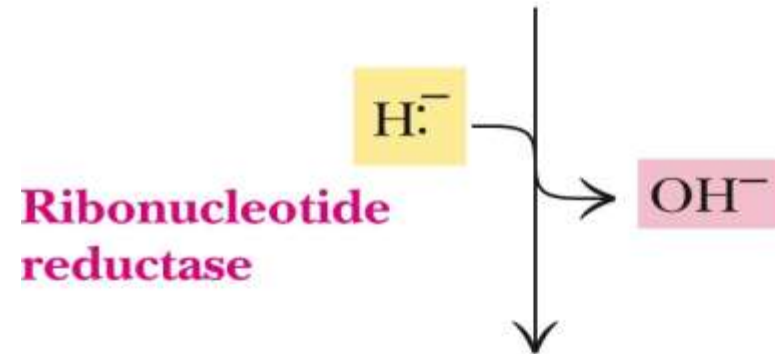
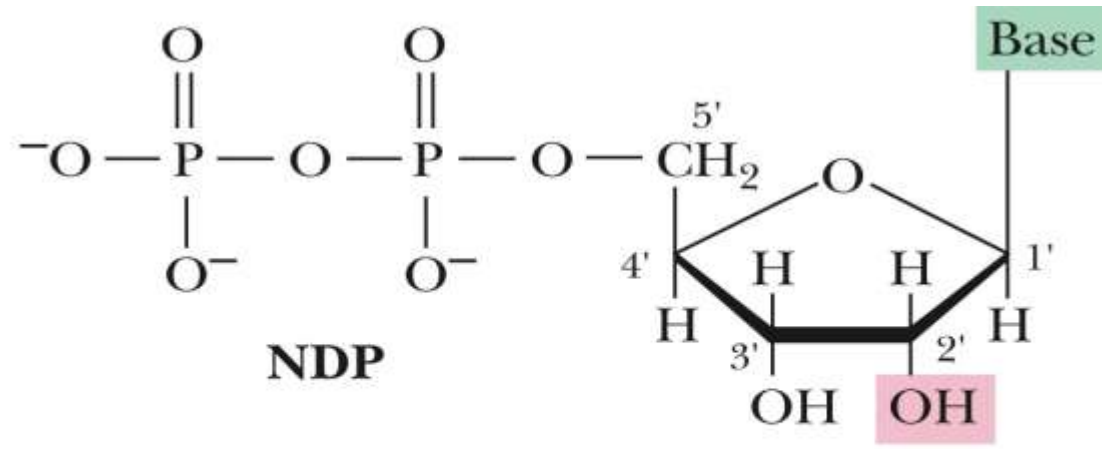


Figure 12
Deoxyribonucleotide synthesis
involves reduction at the 2'-position
of the ribose ring of nucleoside
diphosphates

Regulation of dNTP Synthesis

- The overall activity of **ribonucleotide reductase** must be regulated
All about is regulating the **ribonucleotide reductase**, by substrate availability and end product inhibition
 - **ATP activates**, **dATP inhibits** at the overall activity site
- Balance of the four deoxynucleotides must be controlled
 - dCTP, dATP, dTTP, and dGTP bind at the **specificity site**
 - to regulate the selection of **substrates and the products made**

Salvage Pathway of Purine

✓ **Two phosphoribosyl transferases** are involved in this pathway:

- **Adenosine phosphoribosyl transferase (APRT)**

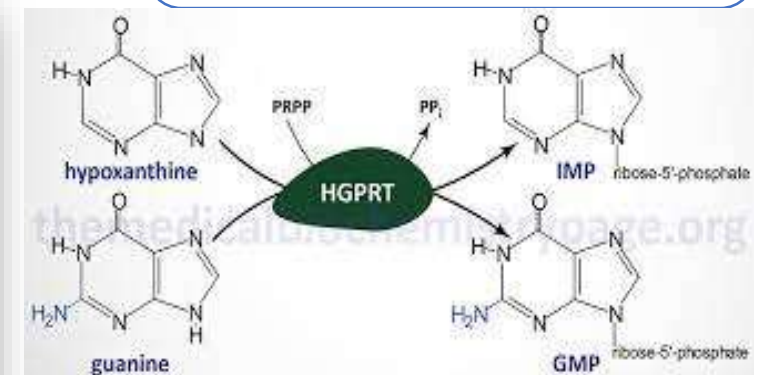
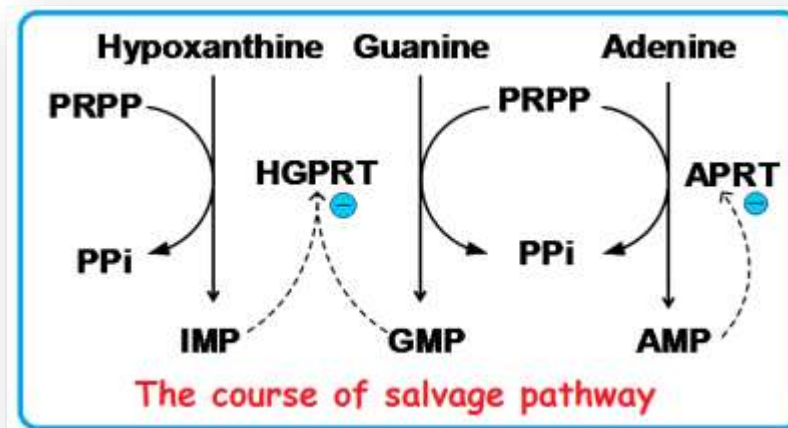


APRT is not very important because it generate little adenine

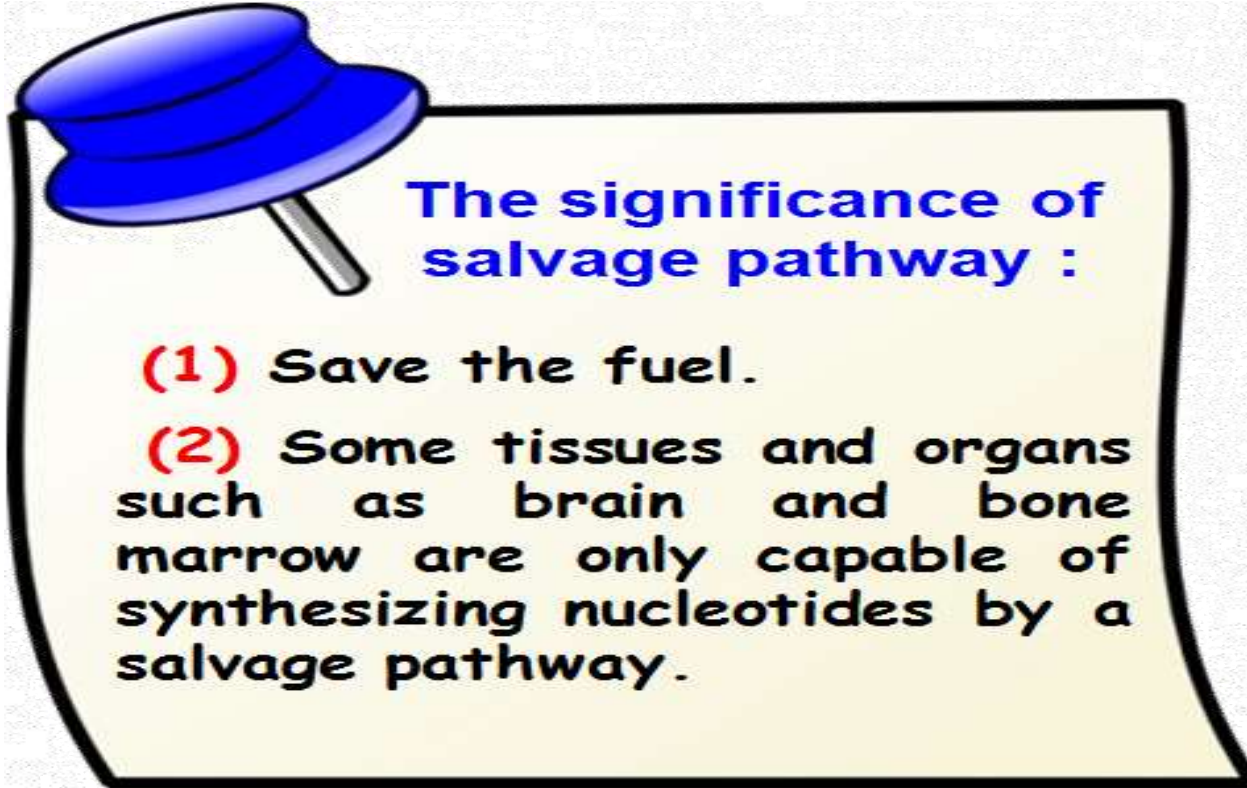
- **Hypoxanthine-guanine phosphoribosyl transferase (HGPRT)**



HGPRT, is exceptionally important and it is **inhibited by both IMP and GMP**



- **Salvage** → **Reusing** the available (**free base** or **nucleoside**) than synthesizing them **from the scratch (de novo)**
- **Two Transferase enzyme** → here **transfers** the **phosphoryl ribose** to the **free bases**
- **AMP** → contains **ribose sugar** attached with **adenine** at **C1** and **1 phosphate** at **C5**
- **IMP (intermediates)** → Synthesis from **Hypoxanthine**
- **APRT** → not highly **regulated**, but **HGPRT** is **highly regulated** by **end-product feedback (GMP and IMP)**



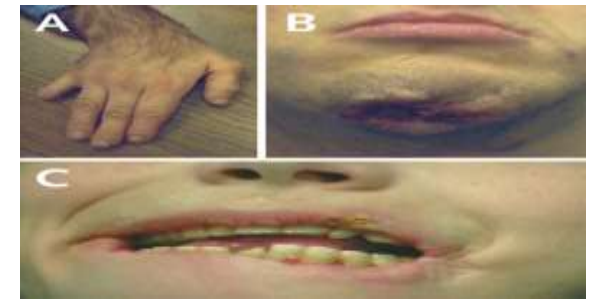
Salvage pathway → used to **recover bases** and **nucleosides** that are formed during the degradation of **RNA** and **DNA** or from an **exogenous source (diet)** to form **nucleotides**

- is important in some organs because some tissues cannot undergo de novo synthesis.
- Bases are synthesized in the liver (de- novo) and transported to RBC (salvaged there)
- Salvage pathways are targets for drug development



Continue...

Biochemistry in medicine



- **Mutations** in **genes** that encode **nucleotide biosynthetic enzymes** can reduce levels of needed nucleotides and can lead to an **accumulation of intermediates**
- **Lesch-Nyhan syndrome** is compulsive **self-destructive behavior**,
 - caused by a **deficiency** of **hypoxanthine-guanine phosphoribosyltransferase (HGPRT)** of salvage pathway,
 - Rare disease is inherited as a **sex-linked recessive disorder**
- In the **absence of HGPRT**,
 - **PRPP levels rise** and **purines** are overproduced by the de novo pathway,
 - resulting in high levels of **uric acid production and gout-like damage to tissue**
- The **brain** is especially **dependent on the salvage pathways**, and
 - this may account for the **central nervous system damage in children with Lesch-Nyhan syndrome**.



Lesch-Nyhan syndrome:

- This syndrome is a rare, X-linked, recessive disorder associated with a virtually complete deficiency of hypoxanthine-guanine phosphoribosyltransferase (HGPRT).
- This deficiency results in an inability to salvage hypoxanthine or guanine, from which excessive amounts of uric acid are produced
- In patients with Lesch-Nyhan syndrome, the hyperuricemia frequently results in the formation of uric acid stones in the kidneys (urolithiasis) and the deposition of urate crystals in the joints (gouty arthritis) and soft tissues.
- In addition, the syndrome is characterized by motor dysfunction, cognitive deficits, and behavioral disturbances that include self-mutilation (biting of lips and fingers)



Figure
Lesions on the lips of Lesch-Nyhan patients caused by self-mutilation.



urine

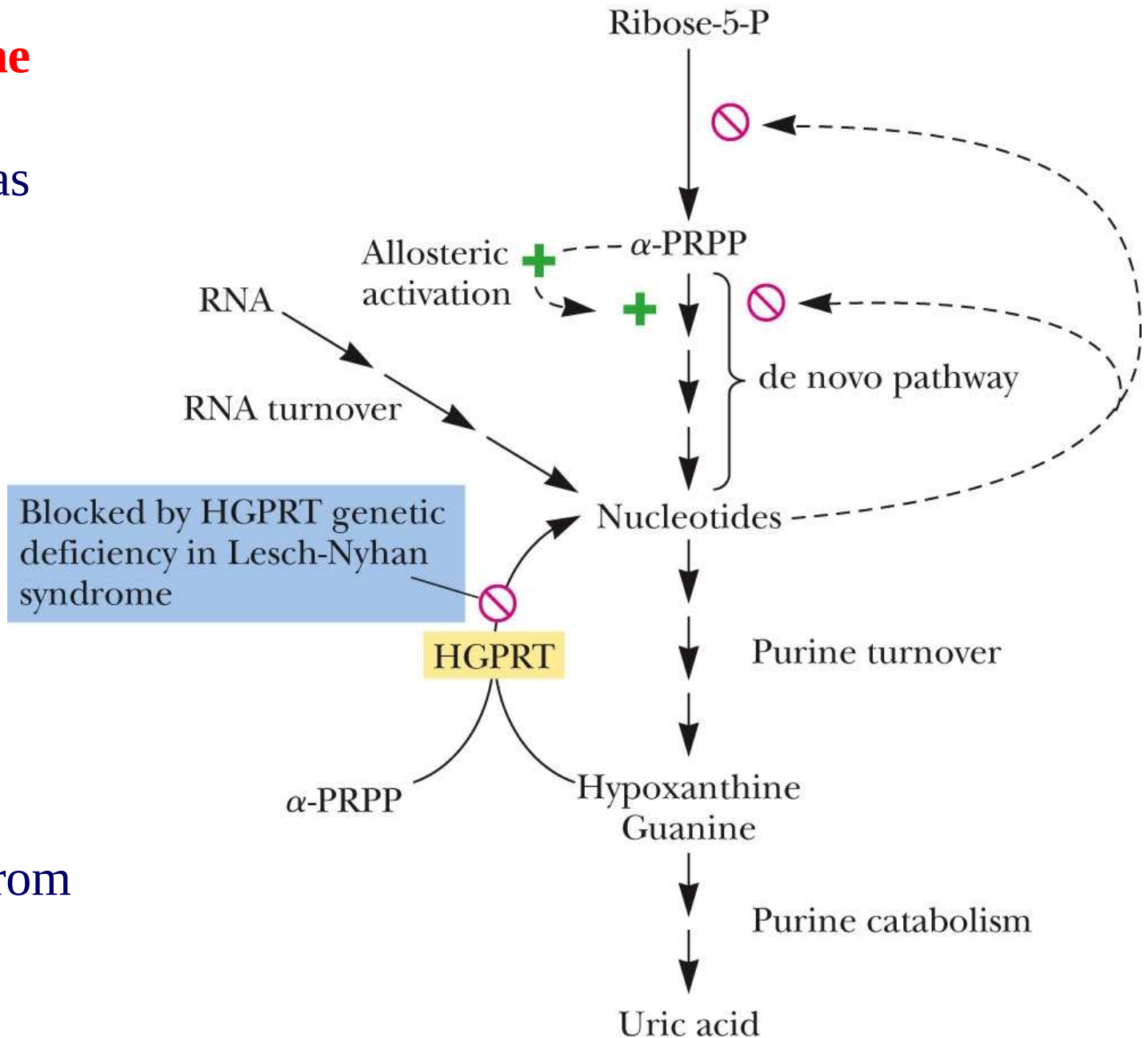
Victims of **Lesch-Nyhan syndrome** experience **severe arthritis** due to accumulation of uric acid, as well as retardation, and other neurological symptoms.

Cause → (HGPRT) Gene **MUTATED**

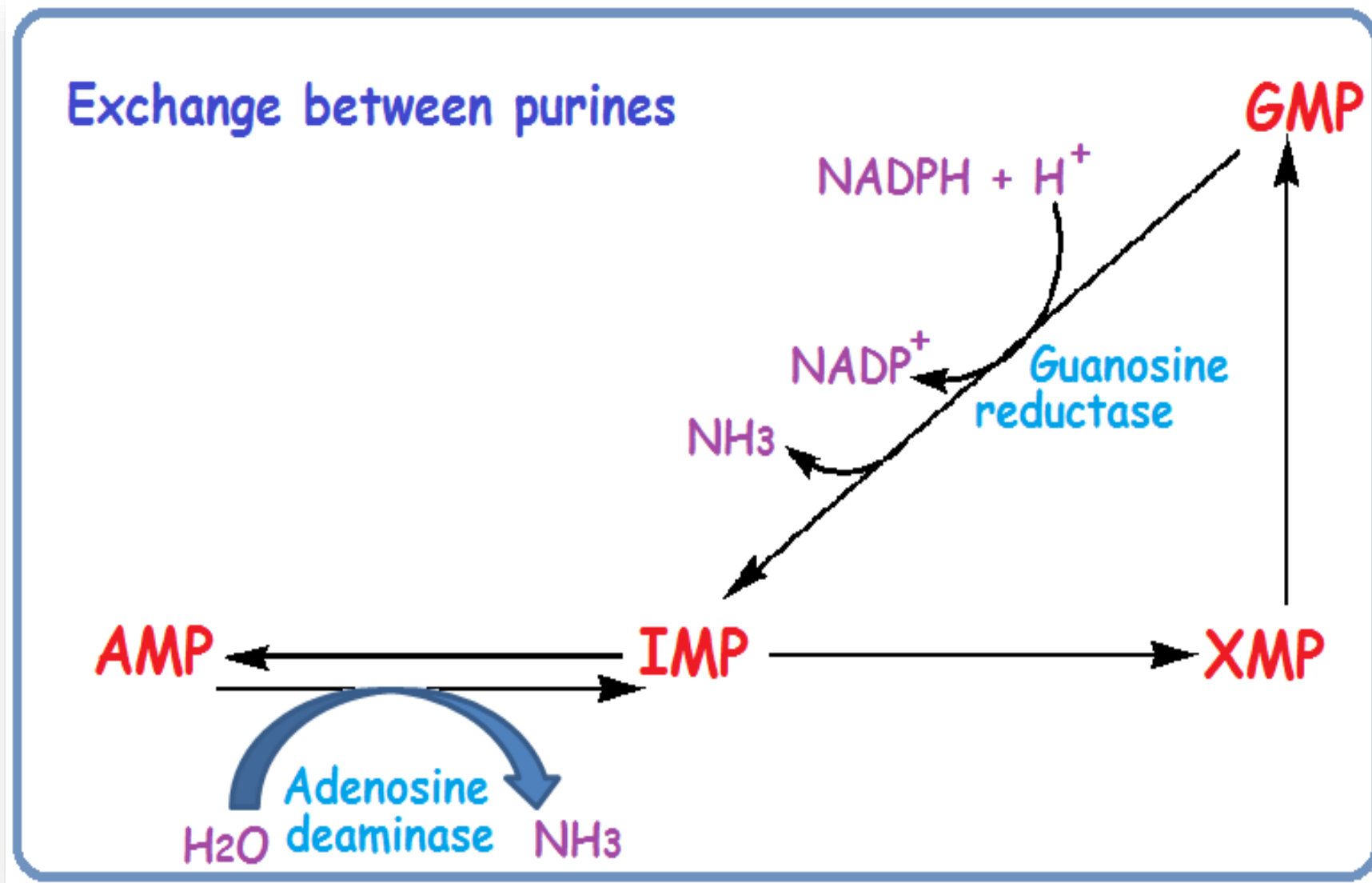
Pathophysiology → IMP and GMP level decrease and increased PRPP level cause stimulation of **De novo- purine synthesis** pathway

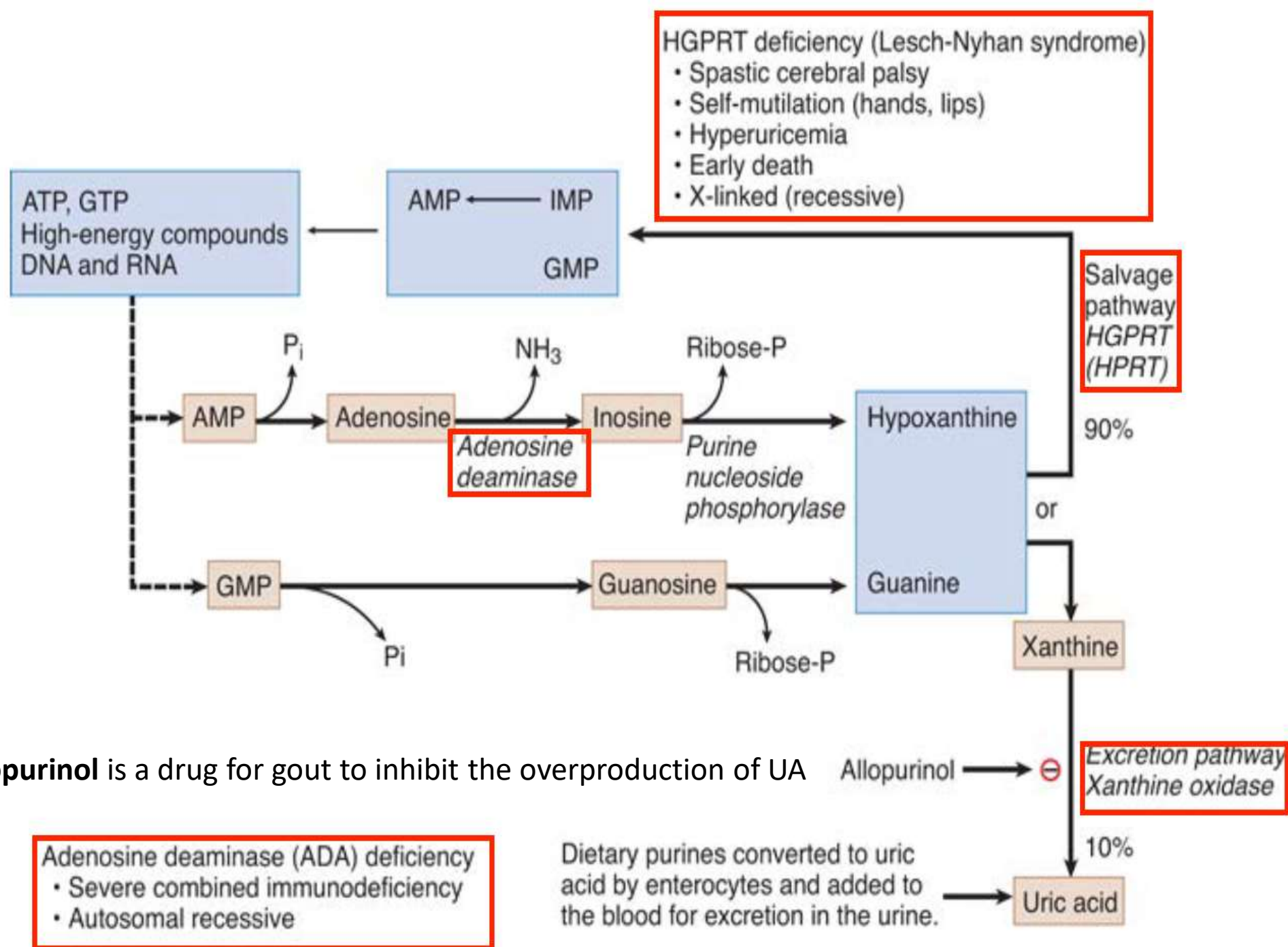
“PRPP High, first precursor (**feed-forward activator**) for **De novo**, **feedback inhibitor** of de novo (IMP and GMP) reduced”

Lesch-Nyhan syndrome results from a **complete deficiency in HGPRT**



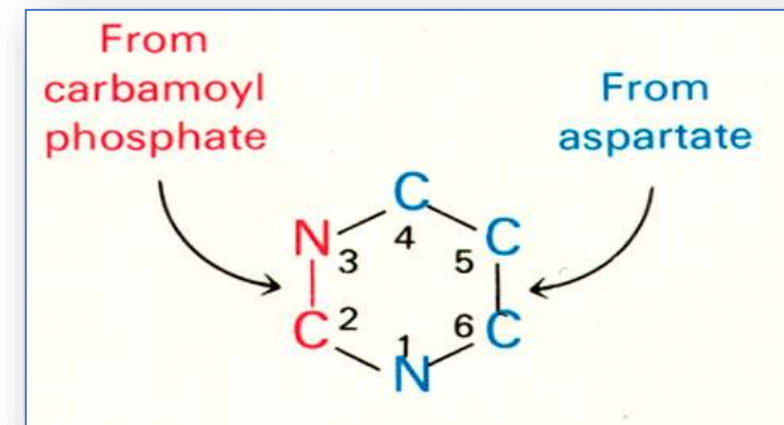
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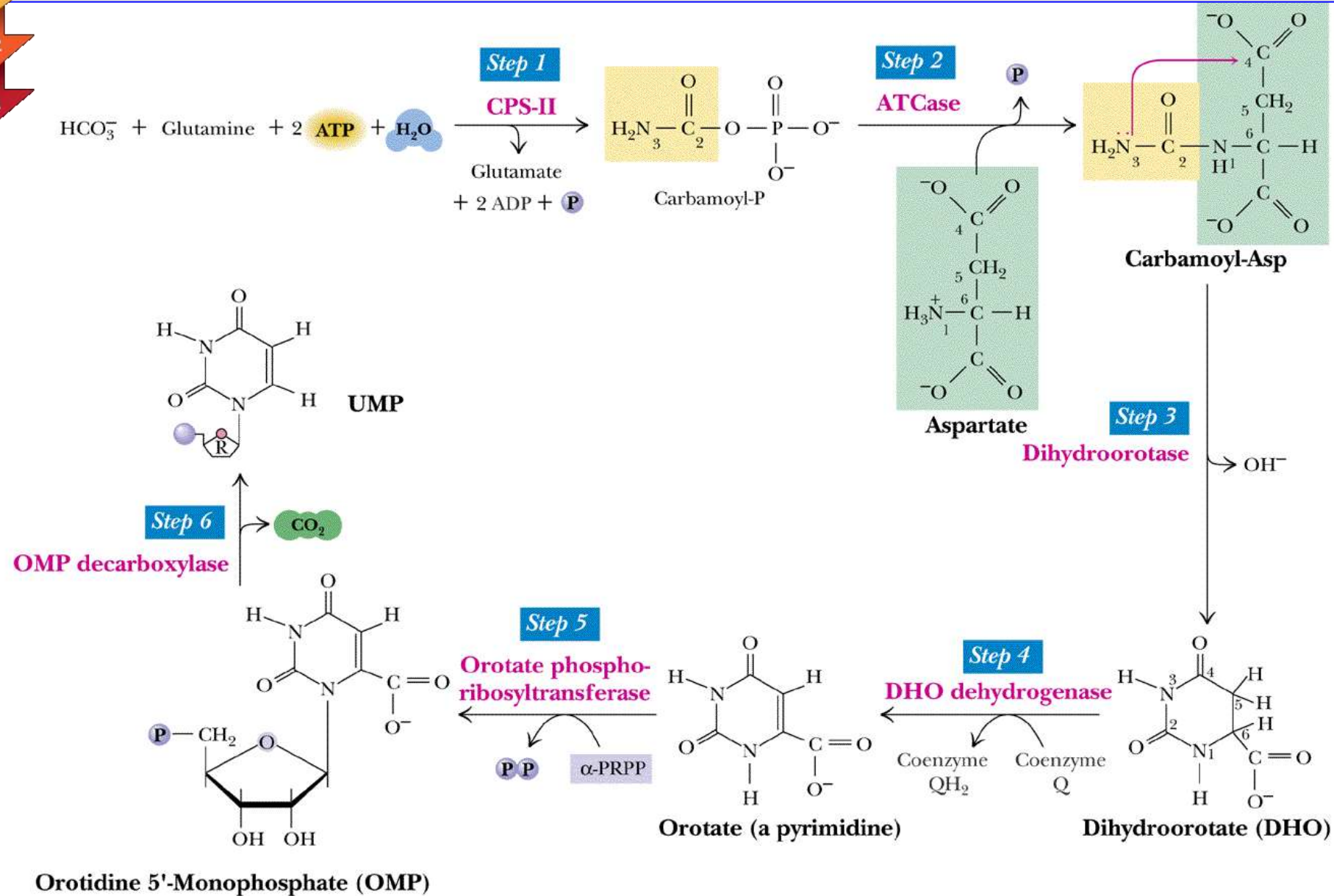


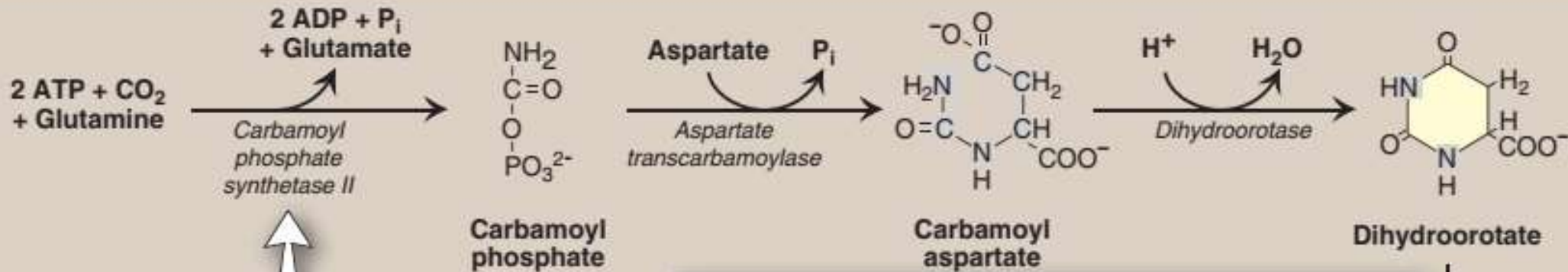
Pyrimidine *De novo* Synthesis

- ✓ It is a **shorter** pathway than purines
- ✓ **The base is made first**, then attached to ribose-P (**unlike purine biosynthesis**)
- ✓ Requires **6 steps** (instead of 11 for purine)
- ✓ The product is **UMP (uridine monophosphate)**
- ✓ **Only 2 precursors** (aspartate and glutamine, plus HCO_3^-) contribute to the 6-membered ring



Pyrimidine *De novo* Synthesis



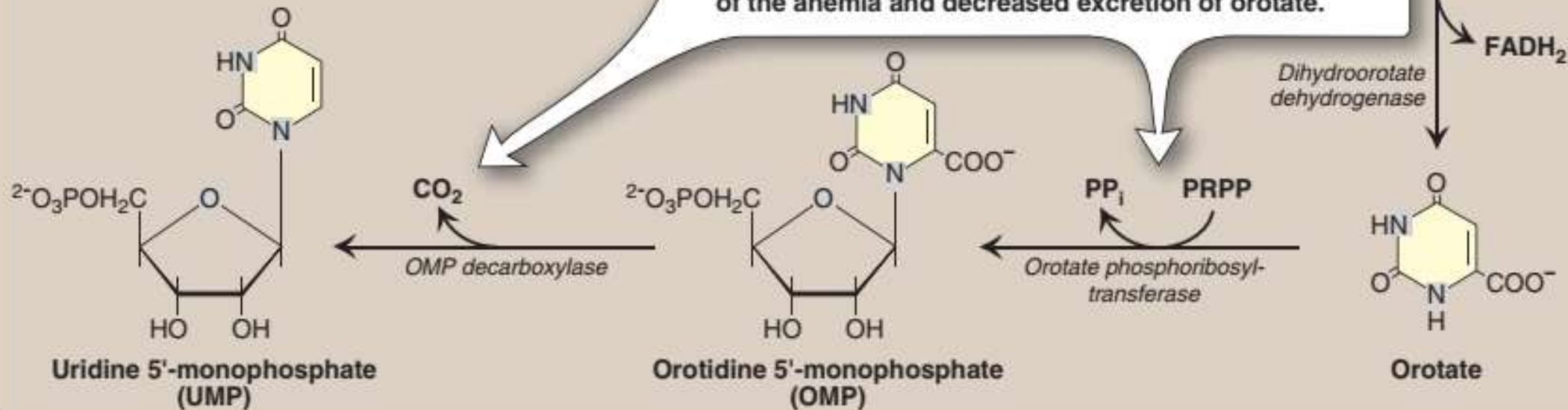


REGULATION OF PYRIMIDINE SYNTHESIS

- In mammalian cells, *carbamoyl phosphate synthetase II* is inhibited by UTP and activated by PRPP.
- In prokaryotic cells, *aspartate transcarbamoylase* is inhibited by CTP and is the regulated step.

OROTIC ACIDURIA

- *Orotate phosphoribosyltransferase* and *OMP decarboxylase* are separate domains of a single polypeptide—*UMP synthase*.
- Low activities of *orotidine phosphate decarboxylase* and *orotate phosphoribosyltransferase* result in poor growth, megaloblastic anemia, and the excretion of large amounts of orotate in the urine.
- Administration of uridine results in improvement of the anemia and decreased excretion of orotate.





Orotic aciduria

- ✓ Because the products of pyrimidine catabolism are soluble, **few disorders** result from excess levels of their synthesis or catabolism
- ✓ Two inherited disorders affecting pyrimidine biosynthesis are the result of **deficiencies in the bifunctional enzyme** catalyzing the last two steps of UMP synthesis, **orotate phosphoribosyl transferase, and OMP decarboxylase.**
- ✓ These **deficiencies result in orotic aciduria** that **causes retarded growth, and severe anemia**

- Orotic acid urea → rare autosomal recessive disorder → An extremely rare disorder that affects both men and women.
- Deficiency of last 2 Bifunctional enzymes UMP synthase (steps 5 and 6) → Elevated Substrate (orotate) appears in the urine (O acid urea) orotic acid
- OAU → cause megaloblastic anemia (defect in RBC development and maturation)
- Treatment → is administering Uridine base *and* blood transfusion for the anemia

Disorders of Pyrimidine Metabolism; Orotic Aciduria

- It is an autosomal recessive disease.

Cause:

- Results from absence of either or both of the enzymes, both orotate phosphoribosyltransferase (OPRTase) and OMP decarboxylase.

It is of 2 types:

1. Type I orotic aciduria:

- Due to genetic disorder of a protein acting as both orotate phosphoribosyltransferase and OMP decarboxylase.
- Orotate fails to be converted to uridylate.

2. Type II orotic aciduria: due to defect of OMP decarboxylase

- ❖ Orotic aciduria may also occur in ornithine transcarbamoylase deficiency (urea cycle enzyme) as carbamoyl phosphate accumulates due to defective conversion to citrulline.

Orotic Aciduria

Features:

- **Poor growth** delays in physical and intellectual development, Neurologic abnormality
- **Megaloblastic anaemia**
- **Does not improve with vitamin B1^{Or} folic acid**
- **Excretion of large amount of orotate in urine**

Treatment:

- **can be successfully treated by feeding cytidine or uridine.**

OTC deficiency

- ↑ orotic acid
- Hyperammonemia
- No megaloblastic anemia



Ornithine transcarbamoylase (OTC)
In Urea cycle

Orotic aciduria

- UMP synthase deficiency
- ↑ orotic acid
- No hyperammonemia
- Megaloblastic anemia



Megaloblastic anemia (macrocytic anemia)

- ✓ An anemia (of macrocytic classification) that results from inhibition of DNA synthesis in red blood cell production.

Nucleotide Mono-, Di-, and Triphosphates Are Interconvertible

- How is the other major pyrimidine ribonucleotide, **cytidine**, formed?
 - ✓ It is synthesized from the **uracil base** of UMP,
 - ✓ **but UMP is converted into UTP before the synthesis can take place**
 - ✓ The di and triphosphates are the **active forms of nucleotides in biosynthesis and energy conversions**

Continue...

- **Nucleoside monophosphates** are converted into nucleoside **triphosphates** in stages.
- **First**, nucleoside monophosphates are converted into diphosphates by specific nucleoside monophosphate kinases that utilize ATP as the phosphoryl-group donor. **For example:**

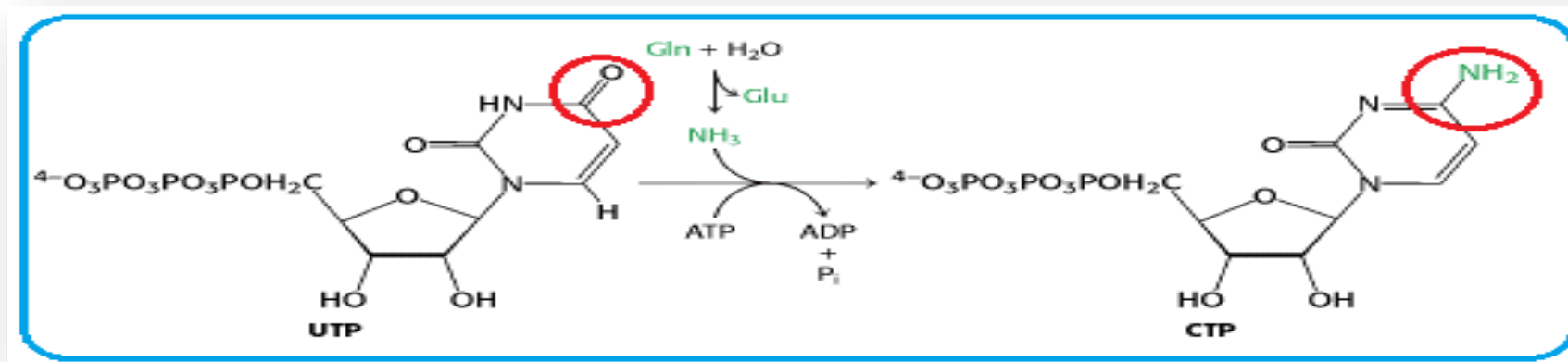


- **Second**, Nucleoside diphosphates and triphosphates are interconverted by nucleoside diphosphate kinase, an enzyme that has broad specificity, in contrast with the monophosphate kinases.



CTP Is Formed by Amination of UTP

- ✓ After **uridine triphosphate** has been formed, it can be transformed into **cytidine triphosphate**
 - ✓ by the **replacement of a carbonyl group by an amino group**.
 - ✓ *By cytidylate synthetase.*
- ✓ This reaction requires **ATP and uses glutamine** as the source of the amino group.
- ✓ CTP can be used then in many biochemical processes, **including RNA synthesis**.



Regulation → GTP acts as an allosteric activator that strongly promotes the hydrolysis of glutamine. This acts to balance the relative amounts of purine and pyrimidine nucleotides. Dysregulation is seen in tumor cell

Thymidylate Is Formed by the Methylation of Deoxyuridylate

- ✓ Uracil, produced by the pyrimidine synthesis pathway, is **not a component of DNA**
- ✓ Rather, DNA contains **thymine**, a methylated analog of uracil
- ✓ Another step is required to generate thymidylate from uracil.
Uracil is RNA, 1st it become **deoxy UMP**,,, then **Thymine (DNA)** formed
- ✓ **Thymidylate synthase catalyzes** this step -deoxyuridylate (dUMP) is methylated to thymidylate (TMP)
- ✓ Require **N⁵, N¹⁰-methylene tetrahydrofolate** (methyl donor) formed from THF and DHF
- ✓ The **methylation** of this nucleotide **facilitates the identification of DNA damage** for **repair** and,
 - ✓ hence, helps preserve the integrity of the genetic information stored in DNA

Continue...

- Conversion of dUMP to dTMP by **thymidylate synthase** and **dihydrofolate reductase**
- In the synthesis of dTMP,
 - all three hydrogens of the added methyl group are derived from the N^5 , N^{10} -methylenetetrahydrofolate, as shown in red and gray

Fluorouracil:

chemotherapeutic agent (anti cancer)

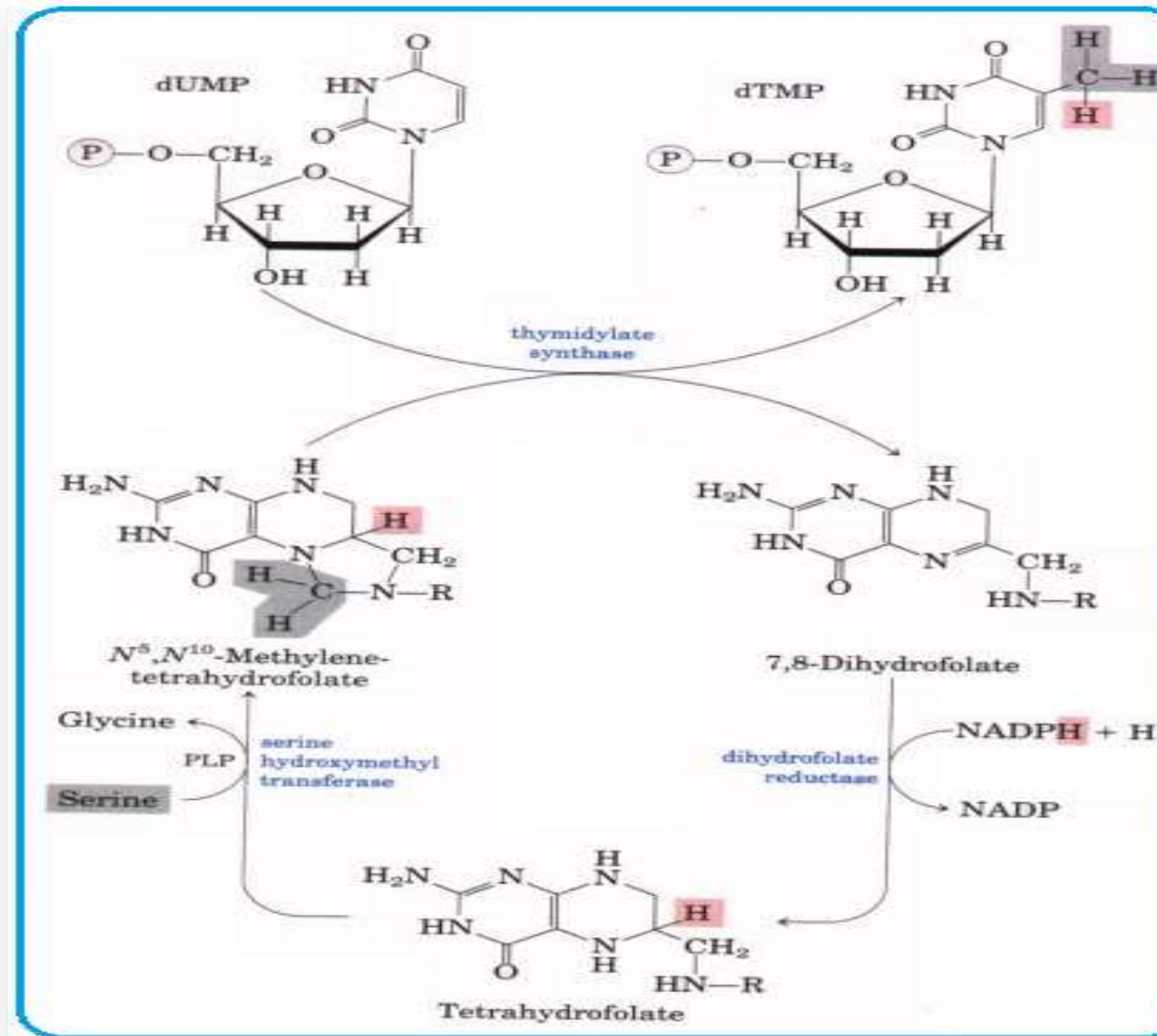
Not an enzyme inhibitor,

it is a suicide inhibitor

It gets converted to FdUMP which binds to and

inactivates thymidylate synthase

Fluorine prevents CH_3 (methyl) incorporation to dUMP by taking the binding site, with no formation of dTMP



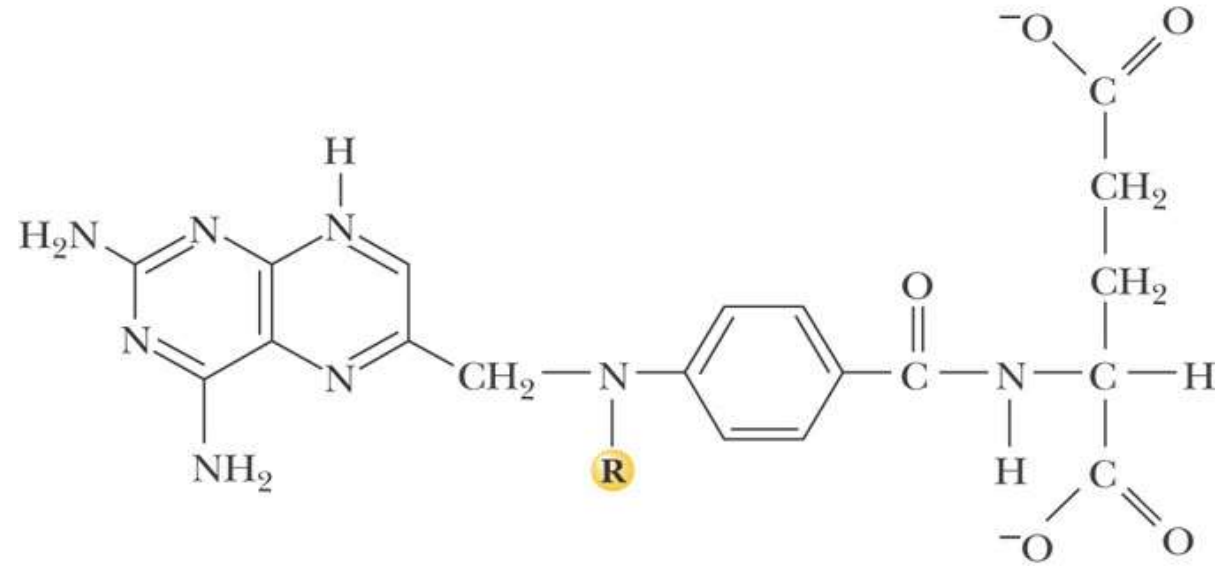
- THF → form of **folic acid (VIT B9)** source of the methyl group “during DHF to N5 N10 MTHF conversion process”
- 5-Fluorouracil (5-FU) chemo drug → *Florine* prevents CH₃ (methyl) incorporation to dUMP by “taking the binding site”, with no **formation of deoxy TMP**
- **5FU** → **chemotherapy** retard the cancer cell proliferation (development and growth), suicide inhibitor (also affect normal cell)
- Pyridoxal phosphate (PLP) → the active form of vitamin B6, is a coenzyme in a variety of enzymatic reactions.
- **Methyl** group from Serine (one carbon cycle) FOLATE CYCLE and SAM cycle

- Precursors and analogs of folic acid employed as **antimetabolites**: **methotrexate**, **aminopterin**, and **trimethoprim**, whose structures are shown here.

- These compounds shown here bind to **dihydrofolate reductase (DHFR)** with about 1000-fold greater affinity than DHF and thus act as virtually **irreversible inhibitors**.

Trimethoprim:

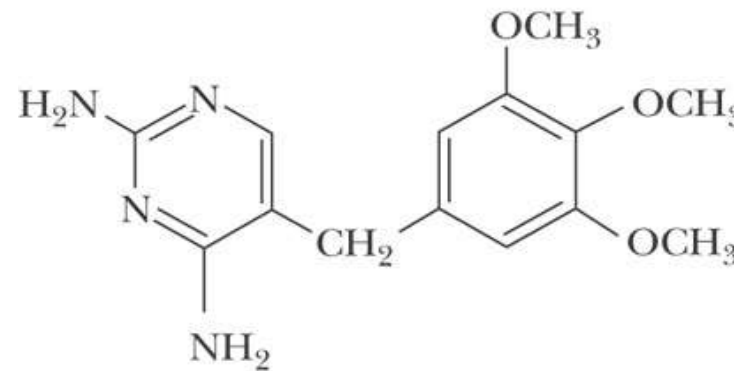
- binds less tightly to mammalian *dihydrofolate reductase* than it does to bacterial *dihydrofolate reductases*.
- Used as an anti-microbial to treat infections
- Anti-bacterial that inhibits THF formation



2-Amino, 4-amino analogs of folic acid

$\text{R} = \text{H}$ Aminopterin

$\text{R} = \text{CH}_3$ Amethopterin (methotrexate)



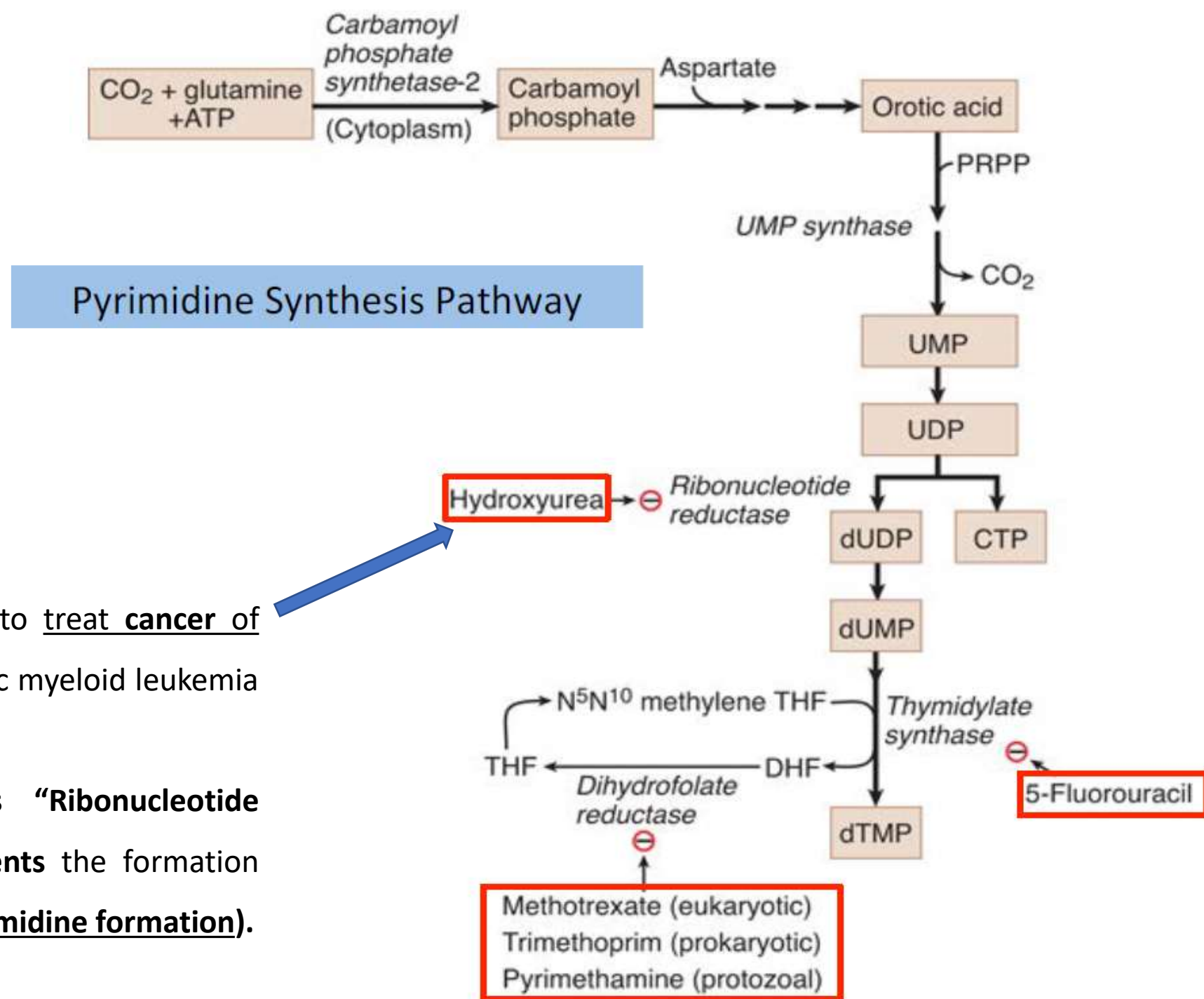
Trimethoprim

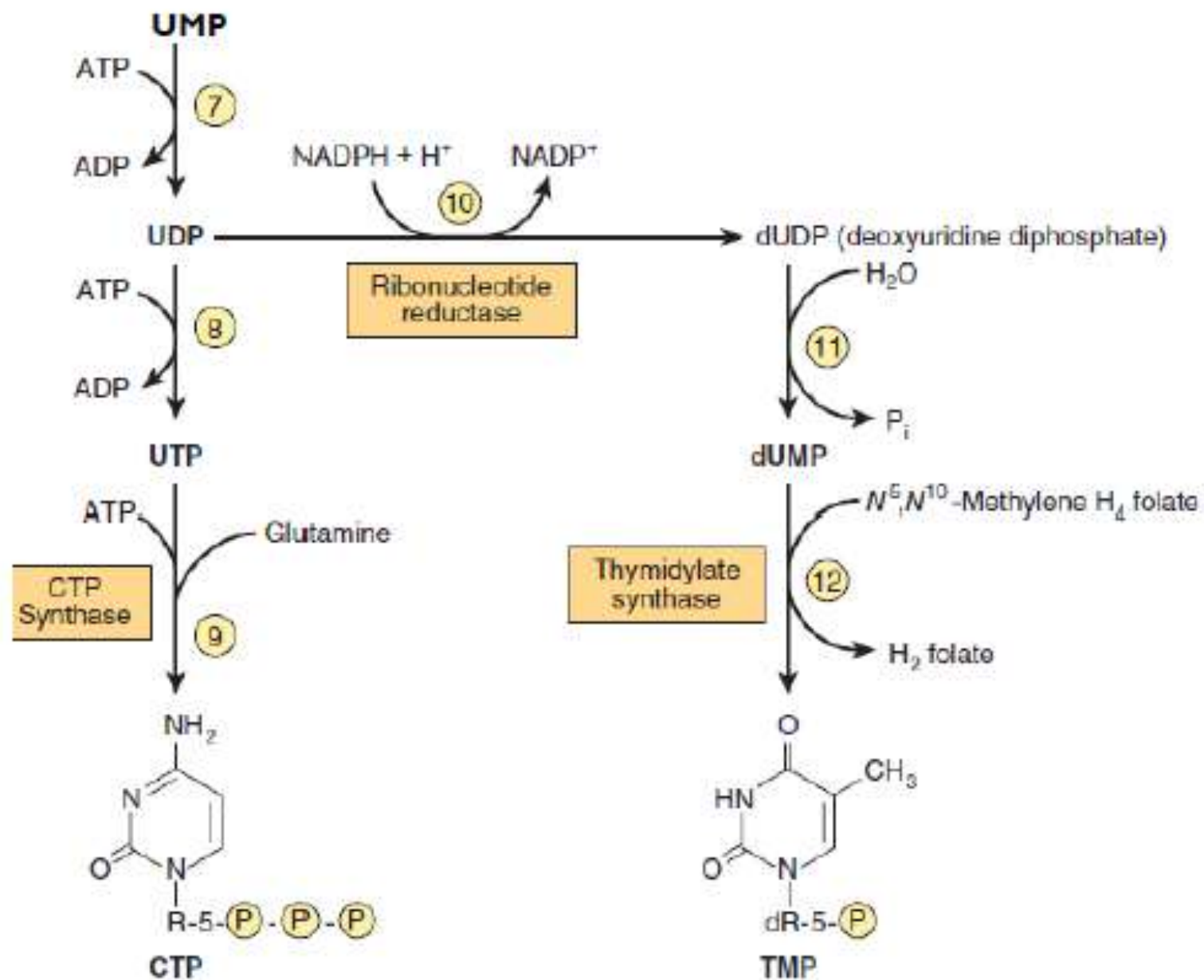
Anti-cancer chemo drugs like **methotrexate** (irreversibly inhibit DHFR enzyme and prevent THF formation from DHF). **Also**, inhibit **Thymidine** formation

“Ribonucleotide reductase” → catalyzes the formation of **deoxyribonucleotides (DNA)** from **ribonucleotides (RNA)** by removing the 2'-hydroxyl group of the **ribose** ring of **nucleoside diphosphates**.

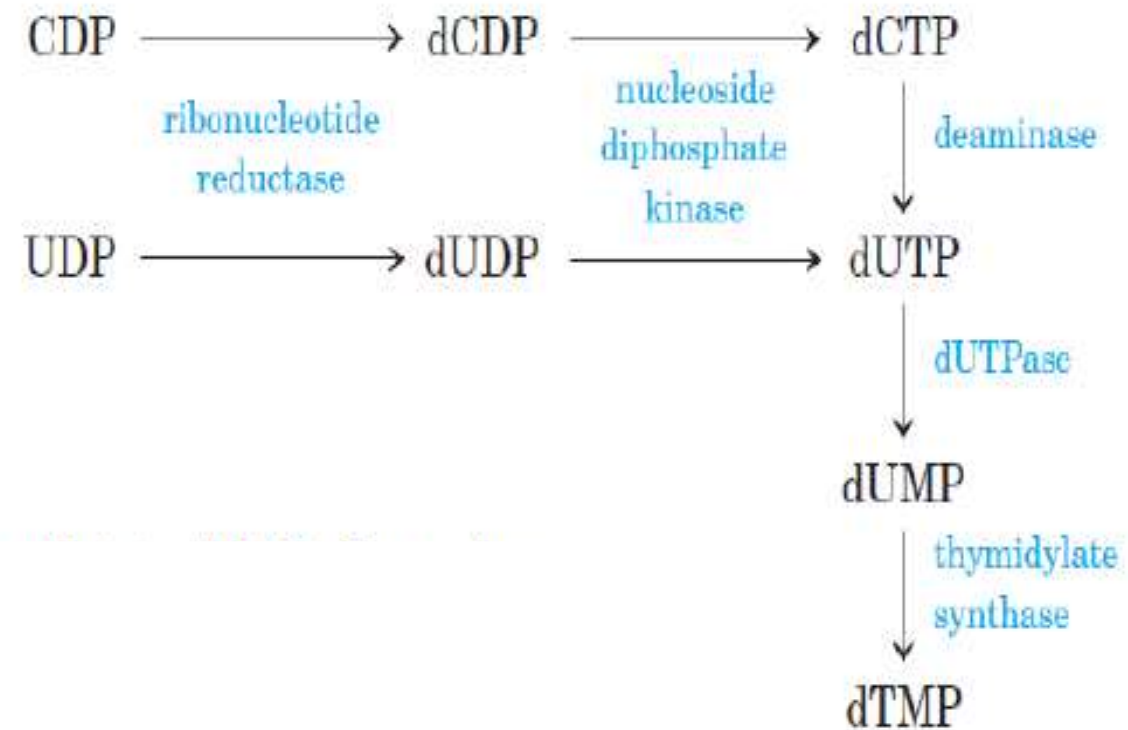
Hydroxyurea is used to treat cancer of the **WBC** called chronic myeloid leukemia (CML).

Hydroxyurea inhibits “**Ribonucleotide reductase**” and **prevents** the formation of **dUDP** (prevent thymidine formation).





Thymidylate Is Derived from dCDP and dUMP



Regulation of pyrimidine synthesis:

- CPSII, aspartate transcarbamylase and dihydroorotase are present as *multienzyme complex*.
- Orotate phosphoribosyl transferase and OMP – decarboxylase are present as *single functional* enzyme. Due to clustering of these enzymes, the synthesis is well coordinated.
- Dihydroorotate dehydrogenase is **mitochondrial** enzyme.

- *(CPSII and aspartate transcarbamoylase)*

And (**OPRTransferase** and **OMP-decarboxylase**) are sensitive to **allosteric regulation**.

- *CPSII is main regulatory enzyme in mammalian cells.*

- **CPS II** - *inhibited by UTP* .

- activated by PRPP

- Aspartate transcarbamoylase :

main regulatory enzyme in prokaryotes.

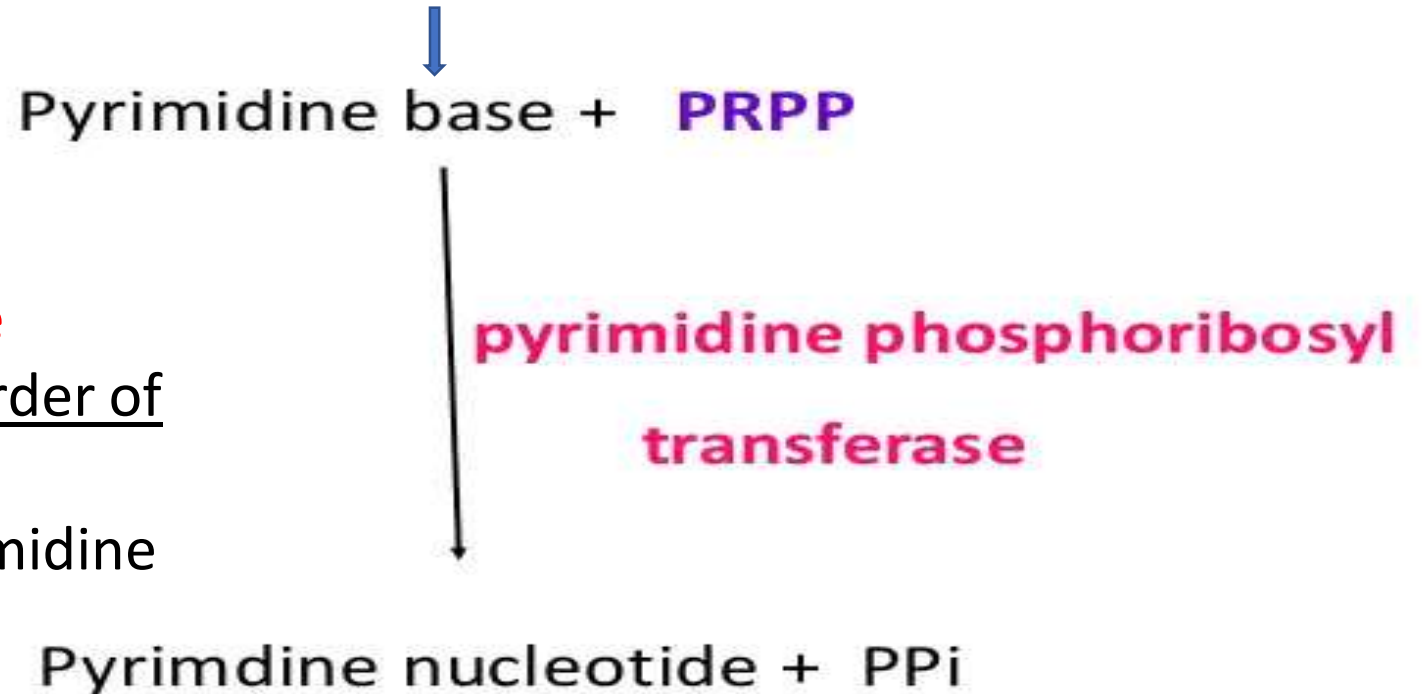
- inhibited by CTP ;activated by ATP

- Requirement of ATP for CTP synthesis and stimulatory effect of GTP on CTP synthase ensures a balanced synthesis of purines and pyrimidines.

GTP allosterically activator for CTP formation (G pair with C)

SALVAGE PATHWAY OF PYRIMIDINE SYNTHESIS

Use the available pyrimidine bases or nucleosides to form a nucleotide



Pyrimidine salvage is effective in the **treatment of orotic aciduria**, a “disorder of pyrimidine nucleotide synthesis”.

Because its alternative path for pyrimidine synthesis during OAU disease

Part II

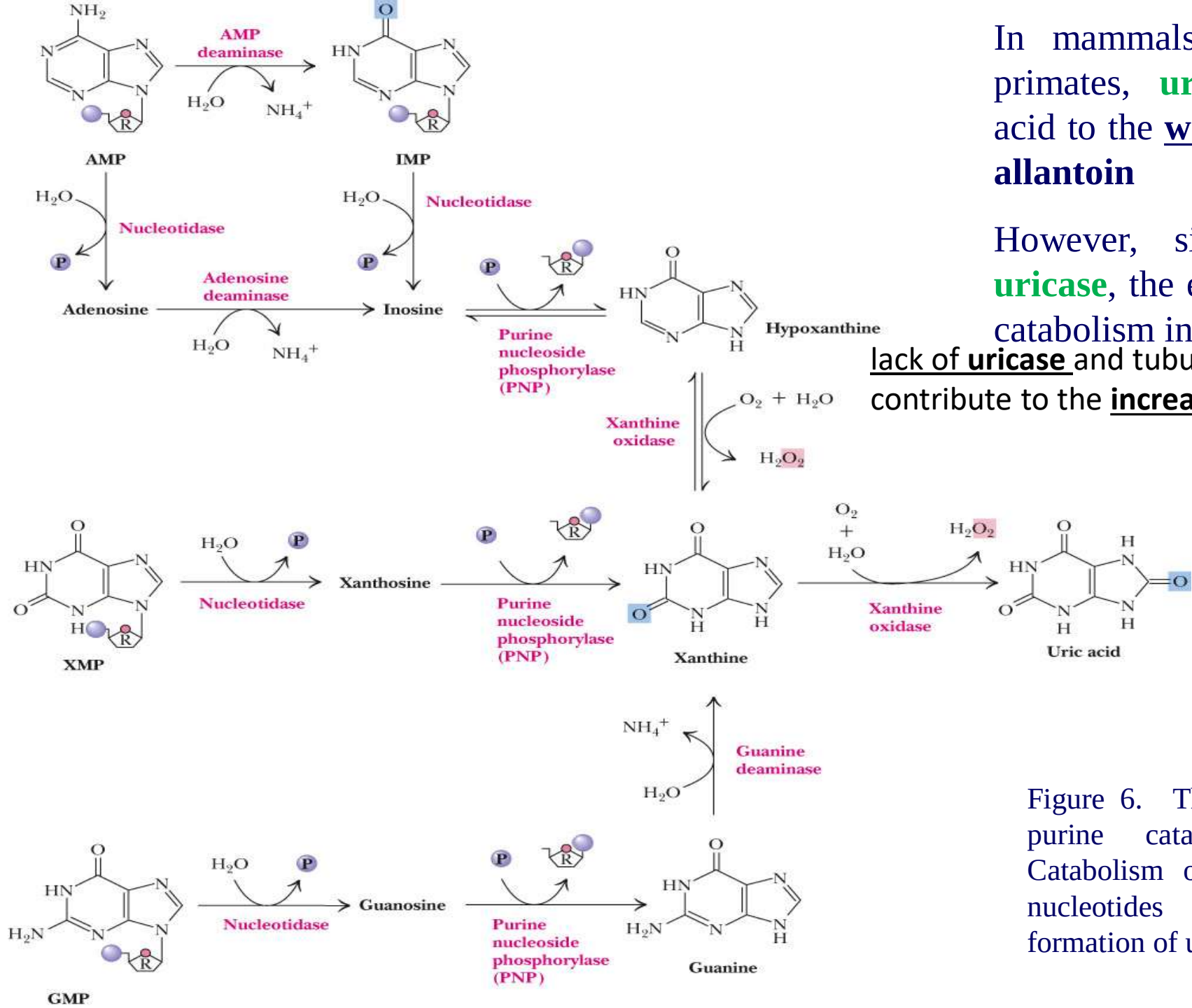
Nucleotides degradation

How Are Purines Degraded?

Purine catabolism leads to uric acid

- **Nucleotidases** and **nucleosidases** release ribose and phosphates and leave free bases
 - **Nucleotidase**: $\text{NMP} + \text{H}_2\text{O} \rightarrow \text{nucleoside} + \text{P}_i$
 - **Nucleosidase**: $\text{nucleoside} + \text{H}_2\text{O} \rightarrow \text{base} + \text{ribose}$
 - **PNP**: $\text{nucleoside} + \text{P}_i \rightarrow \text{base} + \text{ribose-P}$
- The PNP(**purine nucleoside phosphorylase**) products are converted to xanthine by *xanthine oxidase* and *guanine deaminase*
- *Xanthine oxidase* converts xanthine to uric acid
 - Note that xanthine oxidase can oxidize two different sites on the purine ring system
- **Neither adenosine nor deoxyadenosine is a substrate for PNP**
 - Converted to inosine by *adenosine deaminase (ADA)*

Mainly Purine degradation excess end product has **medical consequences** than pyrimidine



In mammals other than higher primates, **uricase** converts uric acid to the water-soluble product allantoin

However, since **humans lack uricase**, the end product of purine catabolism in humans is uric acid. lack of uricase and tubular **reabsorption of UA** by the kidney contribute to the increased uric acid level

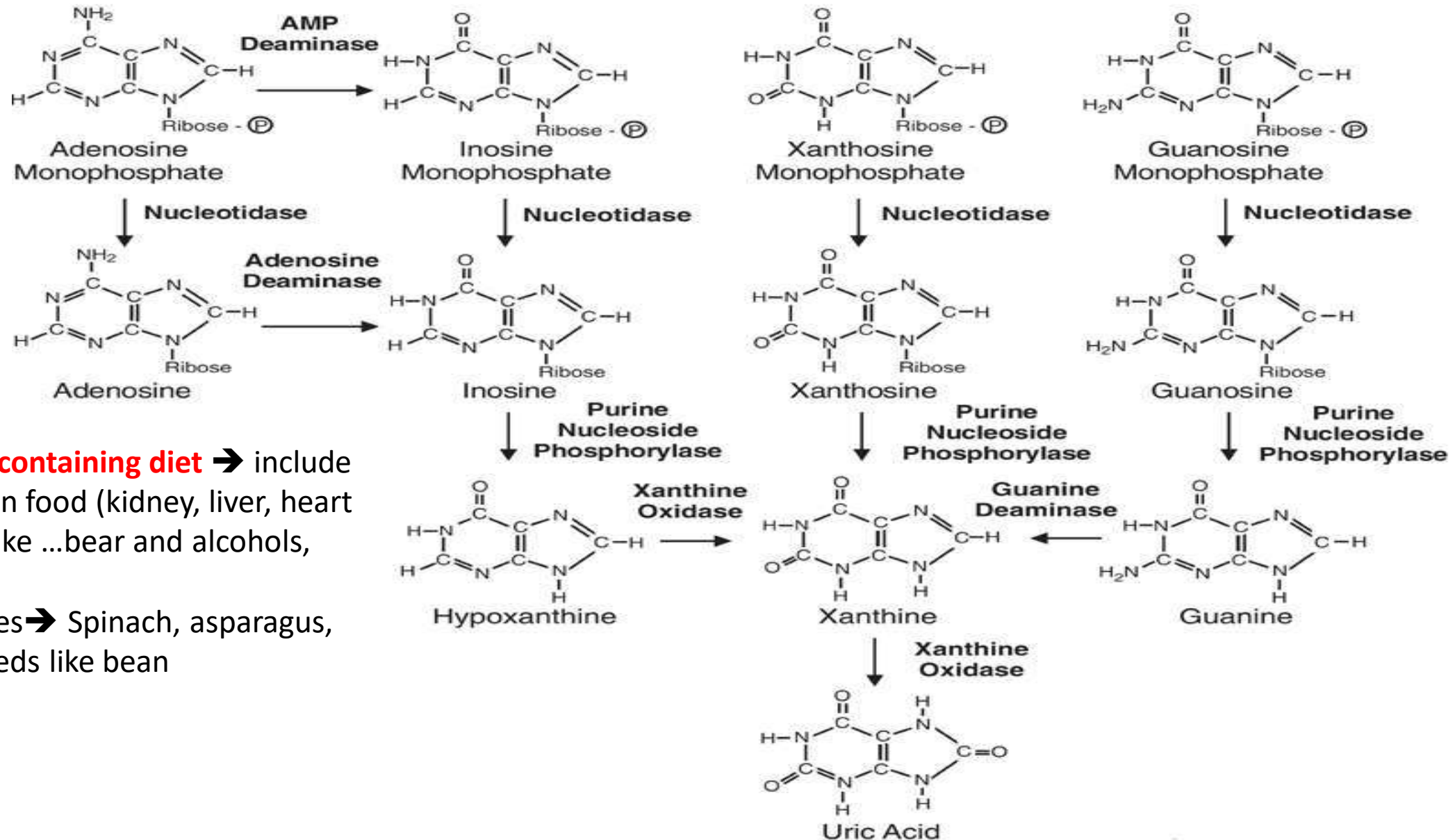
Adult human:

- excreted about 0.6g uric acid per 24 hours
- Comes from dietary purines and from the turnover of purine nucleotides of nucleic acids

Figure 6. The major pathways for purine catabolism in animals. Catabolism of the different purine nucleotides converges in the formation of uric acid

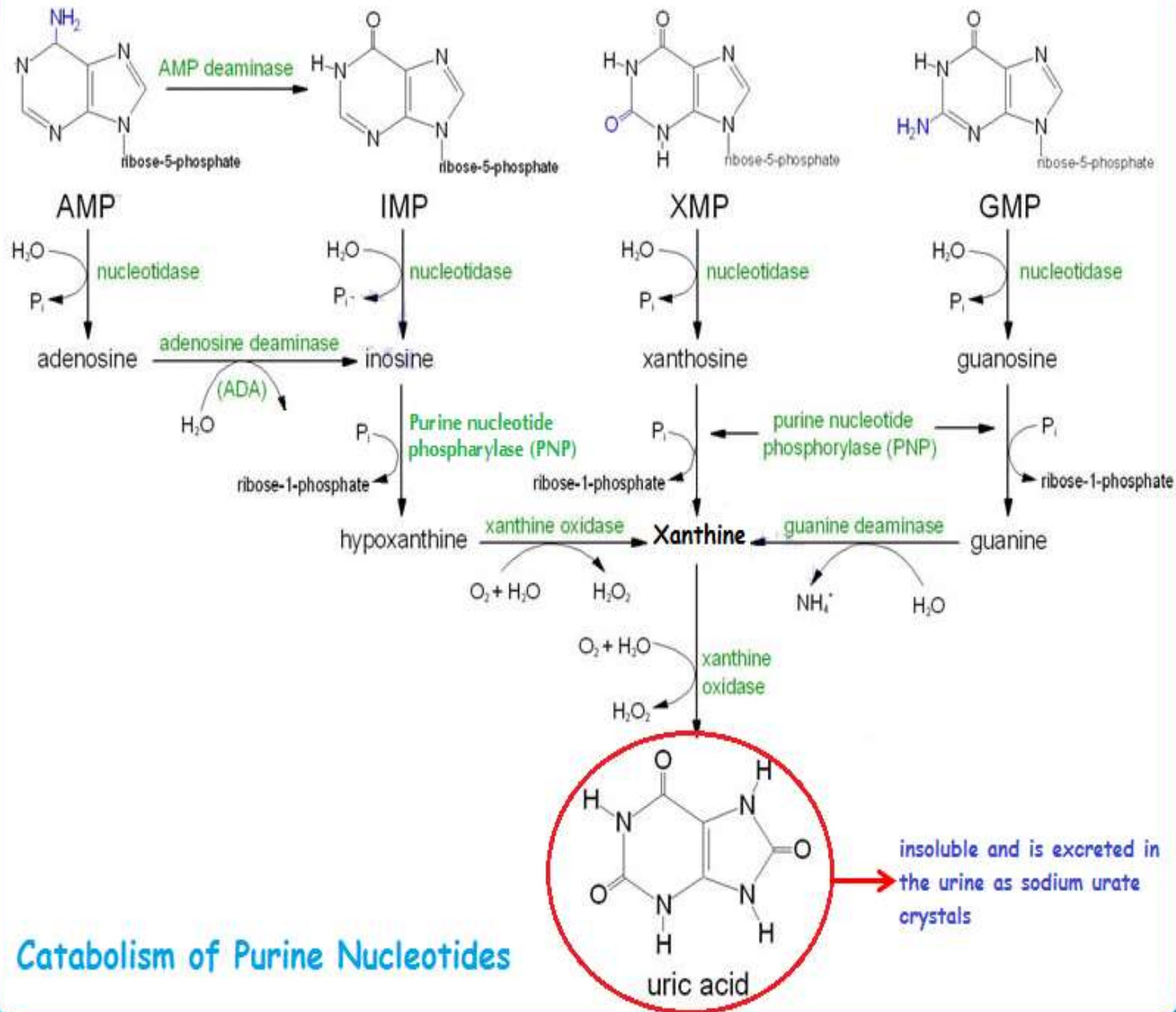
- Nucleotidase → produce nucleoside and PNP further...catalyze the nucleoside
- PNP → produce ribose phosphate and free base, coupling ribose with free **Pi** and forming a free base
- Uricase (urate oxidase) → allantoin is **absent** in **humans**, in urine (extracted) from animals (apes, chimps, monkeys), and as skin moisturizing to remove **warts**, It is frequently present in toothpaste, mouthwash, and other oral hygiene products, **Lipistic** and **shampoo**
- “lack of uricase” and **tubular reabsorption** of **UA** by the **kidney** contribute to the increased uric acid level in human

Biosynthesis of uric acid from purine bases



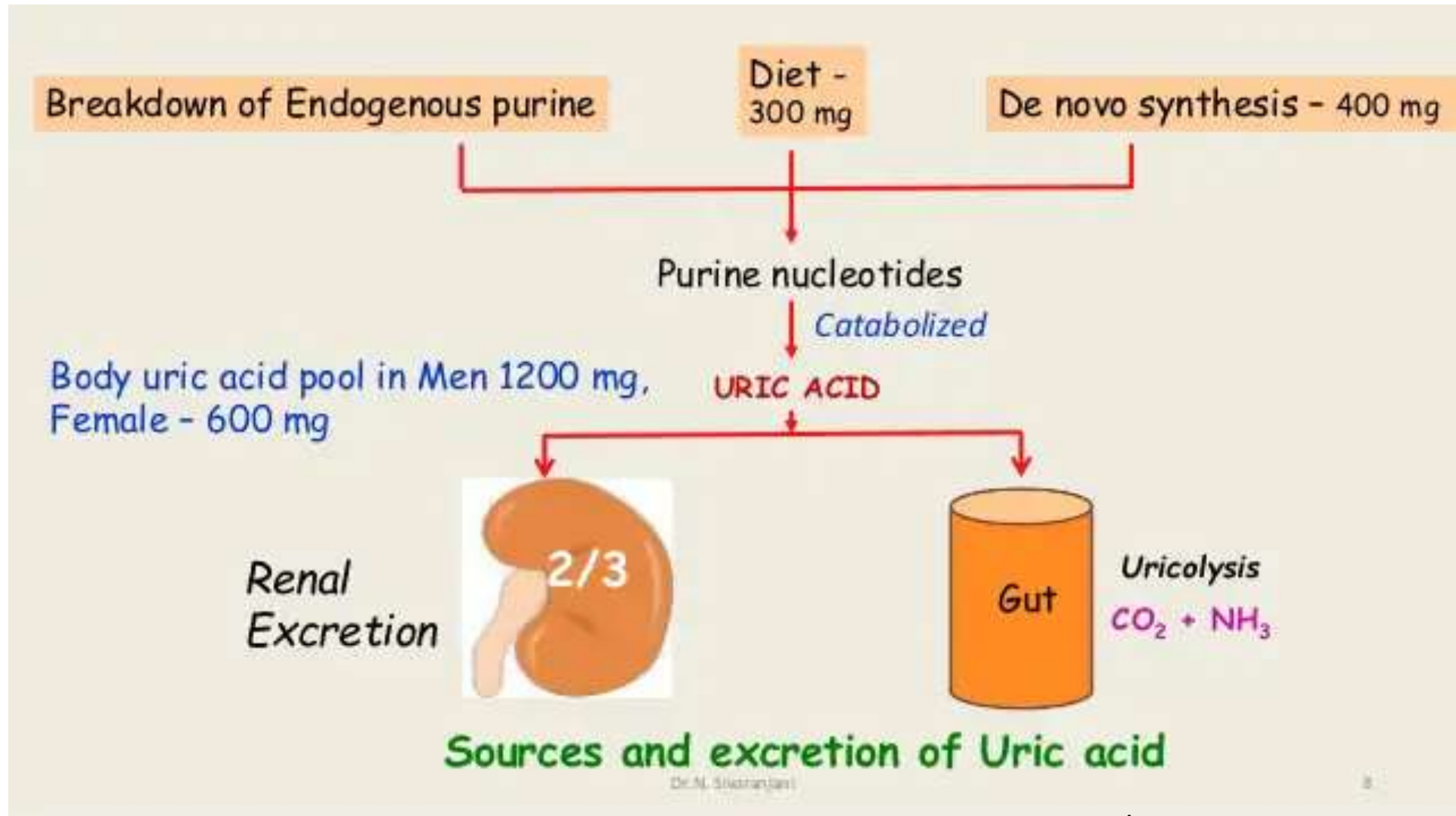
A high purine-containing diet → include red meat, organ food (kidney, liver, heart .., beverages like ...beer and alcohols, soft drinks

From vegetables → Spinach, asparagus, cauliflower, seeds like bean



- Purine nucleotides (AMP and GMP) are degraded by a pathway in which they **lose their phosphate** through the action of **5'-nucleotidase**, to form adenosine and guanosine respectively.

Higher catabolic rates produce **excess uric acids** and develop **arthritis** (accumulation of **sodium urate crystals**)



Uricolysis is the loss of uric acid (1/3) in **sweat** and in **feces**. However, the finding of small insignificant concentrations

Normal serum Uric acid concentration :

- 3 - 7 mg /dl in males , 2 - 5 mg/dl in females
- Uric acid pool -It is on average of 1200 mg

Excretion - 500 to 700 mg /day excreted

- Uric acid is cleared by both glomerular filtration and tubular secretion.
- Uric acid acts as Natural ANTIOXIDANT

Dr. AL. Suleiman (an)

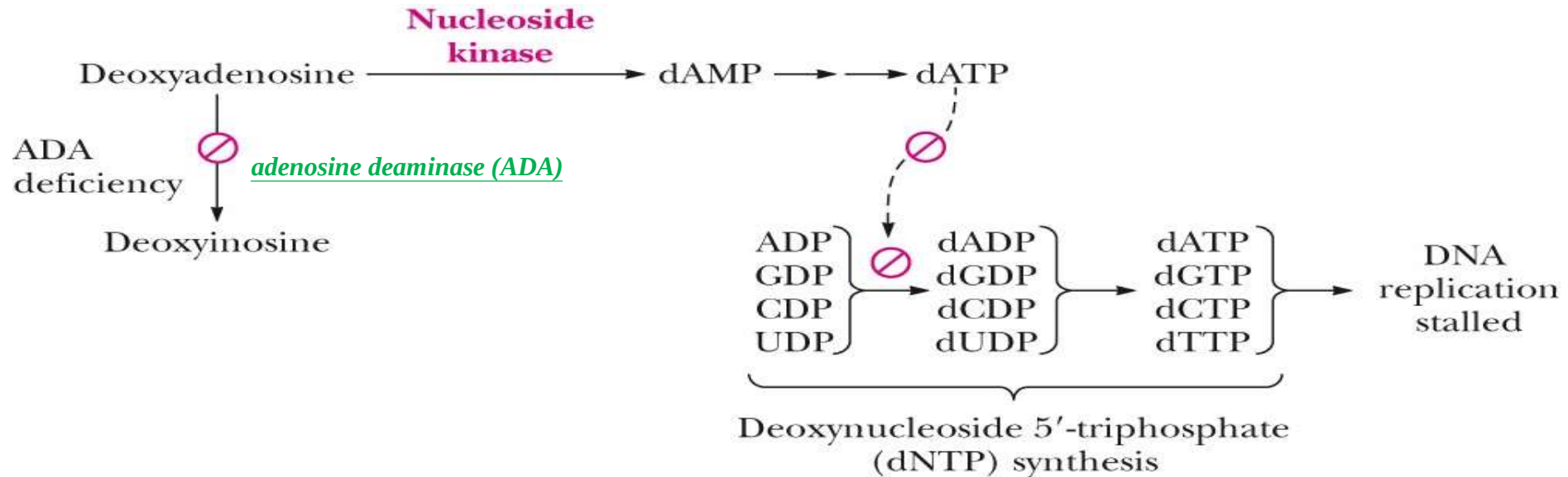
9

Female → higher renal clearance of urate in women, possibly due to their higher plasma estrogen levels.

Uric acid → 50 % of antioxidant activity (importance) in plasma by reducing peroxyl nitrites mediated nitrotyrosine formation

Purine Metabolism Disorders

Severe combined immunodeficiency syndrome (SCID)



The effect of **elevated levels of deoxyadenosine** on purine metabolism.

If **ADA is deficient or absent**, deoxyadenosine is not converted into deoxyinosine as normal.

Instead, it is salvaged by a **nucleoside kinase**, which converts it to dAMP, leading to the accumulation of dATP and **inhibition of deoxynucleotide synthesis**. Thus, **DNA replication is stalled**.

- adenosine deaminase (ADA) deficiency **excess dAMP** and **decreased Inosine**,
“Hypouricemia”
- **The high amount of “deoxy ATP” inhibits** the **“ribo Ntd Reductase”** (stops all deoxy Ntds synthesis) and **stops the replication** process **affects immune cell development (immune deficiency)**, affects **(stops) DNA replication**,
- Cell fell like the **high** product of **d ATP**, causing **inhibition of all deoxy Ntd formation**

Purine Metabolism Disorders...



- ✓ Genetic aberrations in human purine metabolism have been found, some with serious consequences
- ❑ **Severe combined immunodeficiency disease (SCID)**
 - The **deficiency** of **adenosine deaminase** leads to **SCID** in which T lymphocytes and B lymphocytes **do not develop properly** organ is susceptible to infection
 - Lack of ADA leads to a 100-fold **increase** in the cellular concentration of **dATP**, a **strong inhibitor** of **ribonucleotide reductase** involved in deoxynucleotide biosynthesis
 - **The net effect is to inhibit DNA synthesis**

Hypouricemia: Adenosine deaminase (ADA) deficiency:

- A deficiency of ADA results in an accumulation of adenosine, which is converted to its ribonucleotide or deoxyribonucleotide forms by cellular kinases.
- As dATP levels rise, ribonucleotide reductase is inhibited, thus preventing the production of all deoxyribose-containing nucleotides
no production of other dNTPs (Severe) “NO production of DNA”
- Consequently, cells cannot make DNA and divide.
- The dATP and adenosine that accumulate in ADA deficiency lead to developmental arrest and apoptosis of lymphocytes.

- This deficiency causes severe combined immunodeficiency (SCID) involving T-cell and usually B-cell dysfunction
- It is estimated that in the United States, ADA deficiency accounts for approximately 14% of all cases of SCID.
- Treatment requires either bone marrow transplantation (BMT) or enzyme replacement therapy (ERT).
- Without appropriate treatment, ADA deficient children usually die before 2 years of age

ADA deficiency, “Child isolated” from the environment to prevent infection



Treatment → **Enzyme replacement therapy**

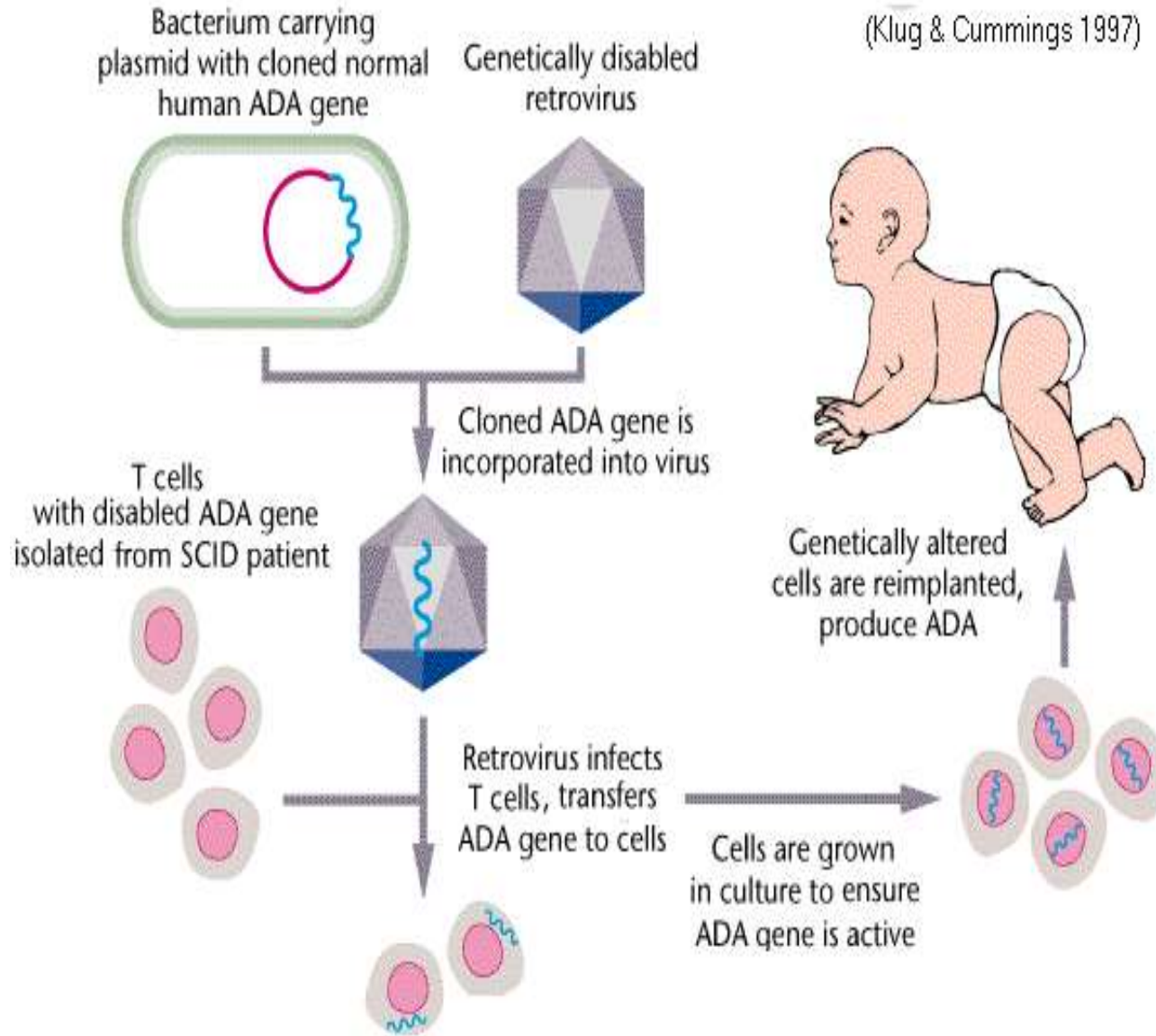
→ **bone marrow transplantation (Containing stem cells)**

- **Hematopoietic** (blood-forming) **stem cell transplantation** is the “**standard treatment**” for **infants** with SCID. Ideally, infants **receive stem cells from a sibling** who is a “**close tissue match**”.

Advanced treatment includes → **gene therapy ...**

SCID Gene therapy

(Klug & Cummings 1997)



- effective treatment for some types of SCID, including X-linked SCID
- In gene therapy, stem cells (T-cells) are obtained from the patient's bone marrow... Then, the normal gene is inserted into the stem cells using a carrier known as a vector, and the corrected cells are returned (implanted) to the patient
- Effective but 1/3 develop leukemia after the treatment (WBC cancer)

Continue...

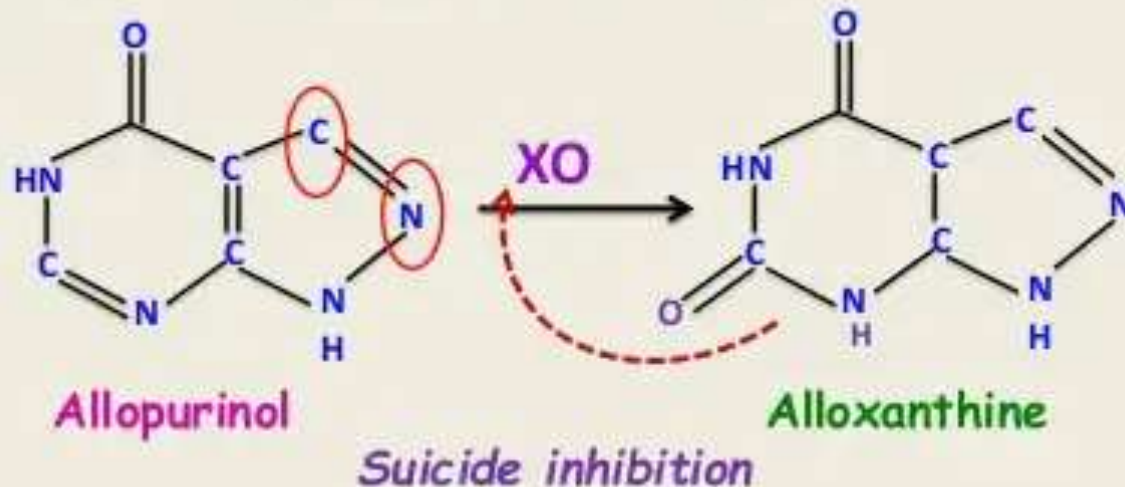


■ Gout

- Is a disease of the **joints** caused by **elevated concentrations of uric acid in the blood and tissues**
- Most forms of gout are the result of **excess purine production (major cause)** and **consequent catabolism** or a **partial deficiency in the salvage enzyme, HGPRT (minor cause)**
- **Allpurinol** is a structural analog of hypoxanthine that strongly **inhibit xanthine oxidase**

3. Xanthine oxidase inhibitors - **ALLOPURINOL**
drug of choice for treatment of Gout.

It is structural analog of hypoxanthine.
Competitively inhibits XO enzyme.



Hypoxanthine , Xanthine
are more soluble and
excreted in urine.

Alloxantin is a **soluble product** of allopurinol

Diseases associated with purine degradation; Gout:

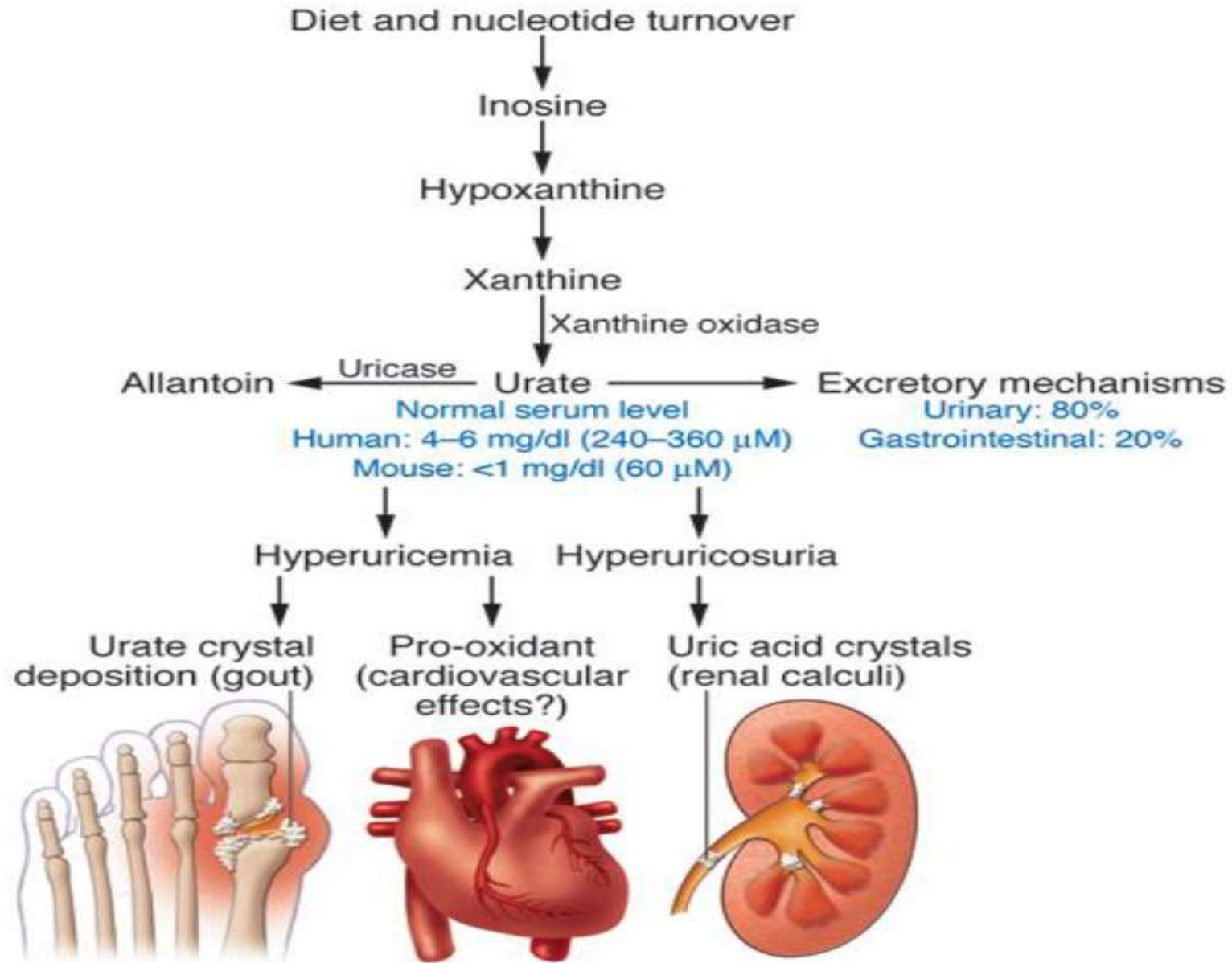
- Gout is a disorder characterized by high levels of uric acid in blood (hyperuricemia)→ as a result of either the overproduction or underexcretion of uric acid.
- The hyperuricemia can lead to **the deposition of monosodium urate crystals** in the joints, and an inflammatory response to the crystals, causing first acute and then progressing to chronic gouty arthritis.
- Nodular masses of monosodium urate crystals (tophi) may be deposited in the soft tissues, resulting in chronic tophaceous gout
- Formation of uric acid stones in the kidney (urolithiasis) may also be seen.

Nodular mass formation (in areas with low temperature) in patients, due to **crystal accumulation in soft tissue**



Figure
Tophaceous gout.







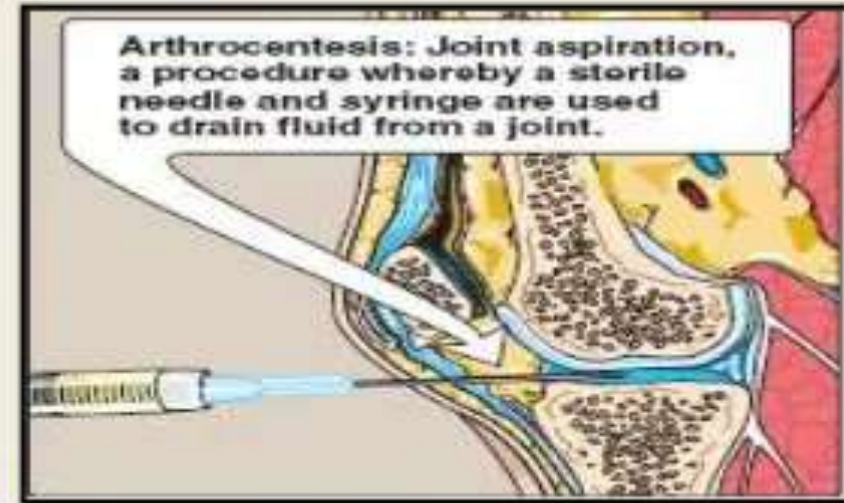
Often HISTORY is - patient have few drinks at night , go to sleep symptomless , but are awakened during early hrs by severe joint pains.

- Hyperuricemia doesn't always cause gout. Over the course of yrs, sharp urate crystals build up in the synovial fluid of the joints.

Precipitating event - infection, surgery, stress or often heavy ALCOHOL drink.

Investigations :

- Serum uric acid level -increased
- Microscopic Examination of Synovial fluid reveals uric acid crystals - rod / needle -shaped crystals.
- Birefringent crystals under polar microscope is diagnostic.



Uric acid crystals in the urine

Management of Gout

- By reducing dietary purine intake and restricting alcohol.
- By increasing renal excretion of urate by uricosuric drugs, which decrease the reabsorption of uric acid from kidney tubules, e.g. probenecid
- By reducing urate production by allopurinol, which is an analogue of hypoxanthine.
- Allopurinol is a competitive inhibitor of xanthine oxidase thereby decreasing the formation of uric acid.
- Xanthine oxidase converts allopurinol to alloxanthine. It is a more effective inhibitor of xanthine oxidase. This is a good example of '**suicide inhibition**'
- Antiinflammatory agents like Colchicine, steroidal drugs such as prednisone, and nonsteroidal drugs such as indomethacin are used to treat Acute attacks of gout

Treatment of gout :

- Low intake of Purine diet- like red meat, acidy fruits and vegetables, lentils
- Restrict Alcohol
- Consume plenty of Water



Drugs:

1. Anti-inflammatory drugs - Colchicine is used for treatment of gouty arthritis. NSAID - indomethacin , ibuprofen. Corticosteroids also useful for acute attacks.
2. Uricosuric drugs - Probenecid.
Uricosuric drugs, which facilitate uric acid transportation are drugs that promote the excretion of uric acid

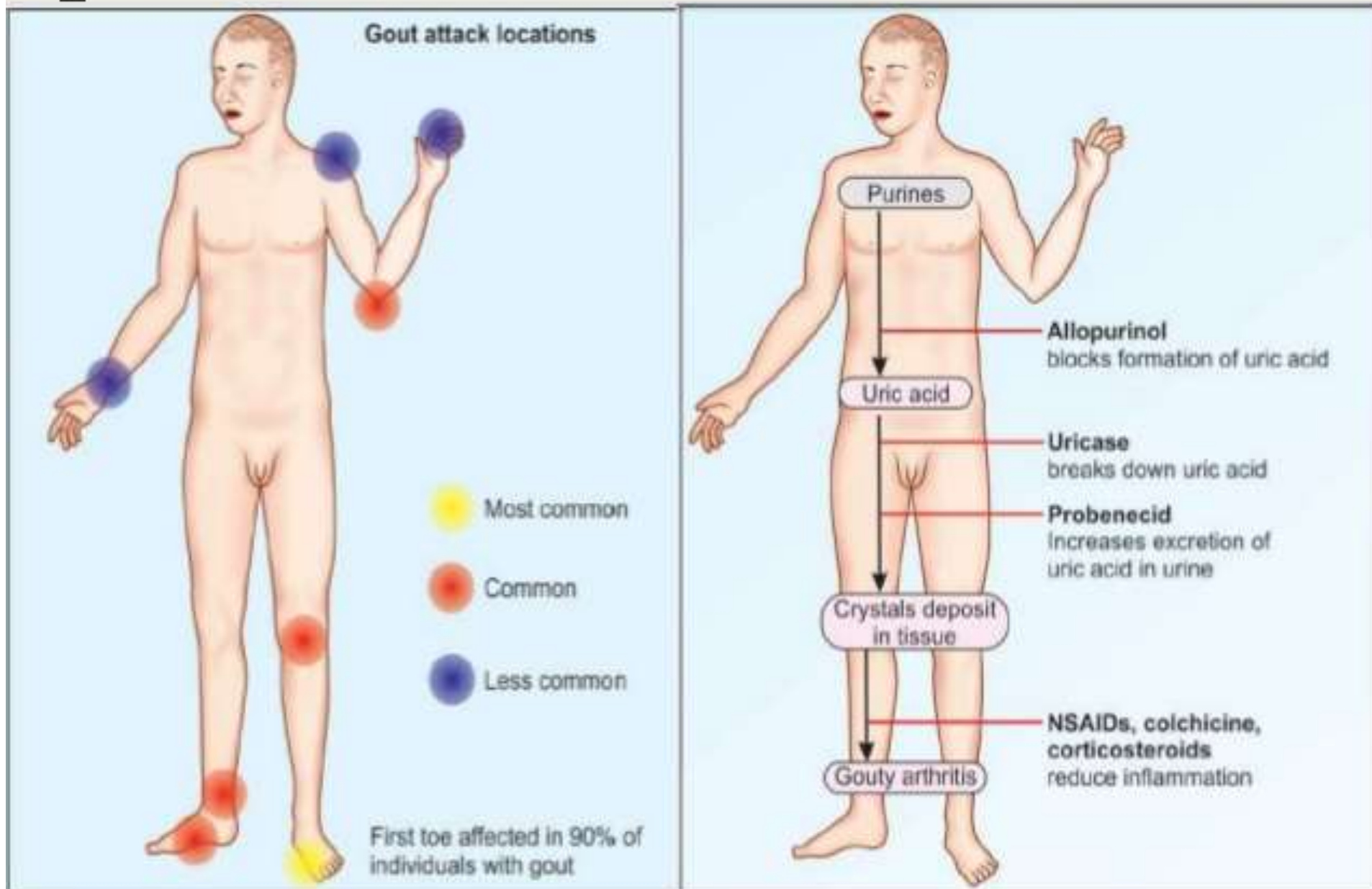


Fig. Location of gout attack

Treatment of GOUT

Acute attack

Colchicine

Steroids

Indomethacin

Longterm therapeutic strategy

Allopurinol

Febuxostat—Nonpurine inhibitor of Xanthine oxidase

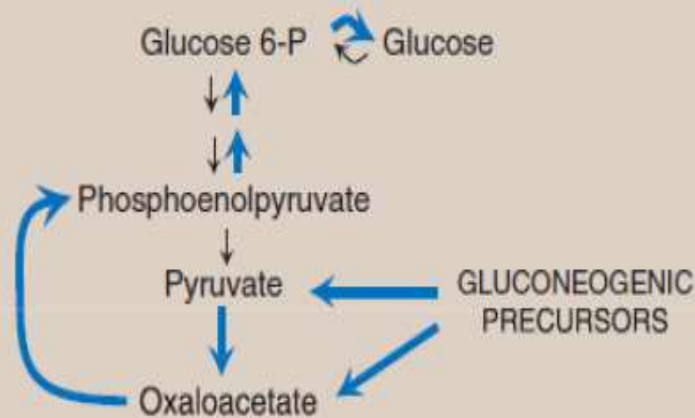
Diet: Avoid diet rich in nucleic acid

Diet recommendation for gout patients

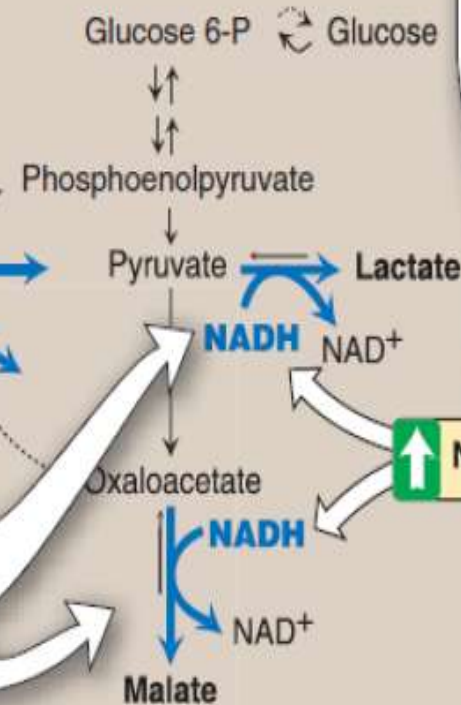
High levels of purines Best To Avoid	Moderate levels of purines Eat Occasionally	Low levels of purines No Restriction
Liver	Asparagus	Carbonated drinks
Kidney	Beef	Coffee
Anchovies	Bouillon	Fruits
Sardines	Chicken	Breads
Herrings	Crab	Grains
Mussels	Duck	Macaroni
Bacon	Ham	Cheese
Scallops	Kidney beans	Eggs
Cod	Lentils	Milk products
Trout	Lima beans	Sugar
Haddock	Mushrooms	Tomatoes
Veal	Lobster	Green vegetables
Venison	Oysters	
Turkey	Pork	
Alcohol esp beer	Shrimp	
	Spinach	

Alcohol precipitates gout

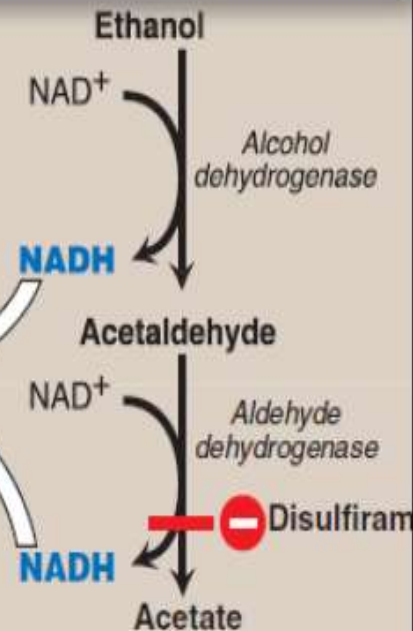
A No ethanol consumed



B Ethanol consumed



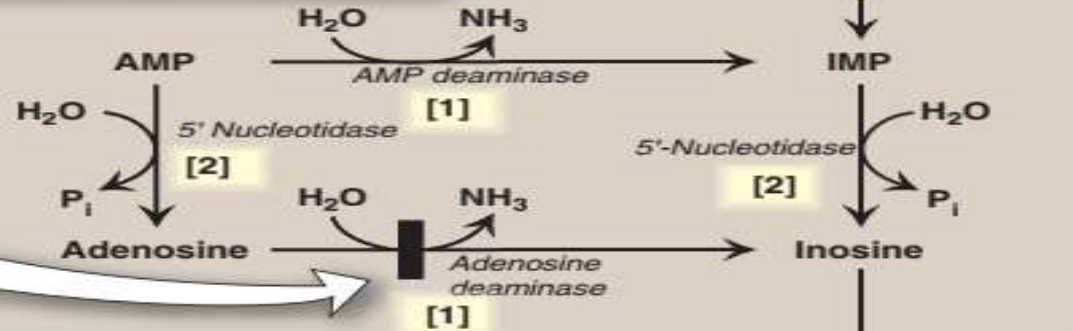
1 The metabolism of ethanol results in a massive increase in the concentration of cytosolic NADH in the liver.



2 The ethanol-mediated increase in NADH causes the intermediates of gluconeogenesis to be diverted into alternate reaction pathways, resulting in decreased synthesis of glucose.

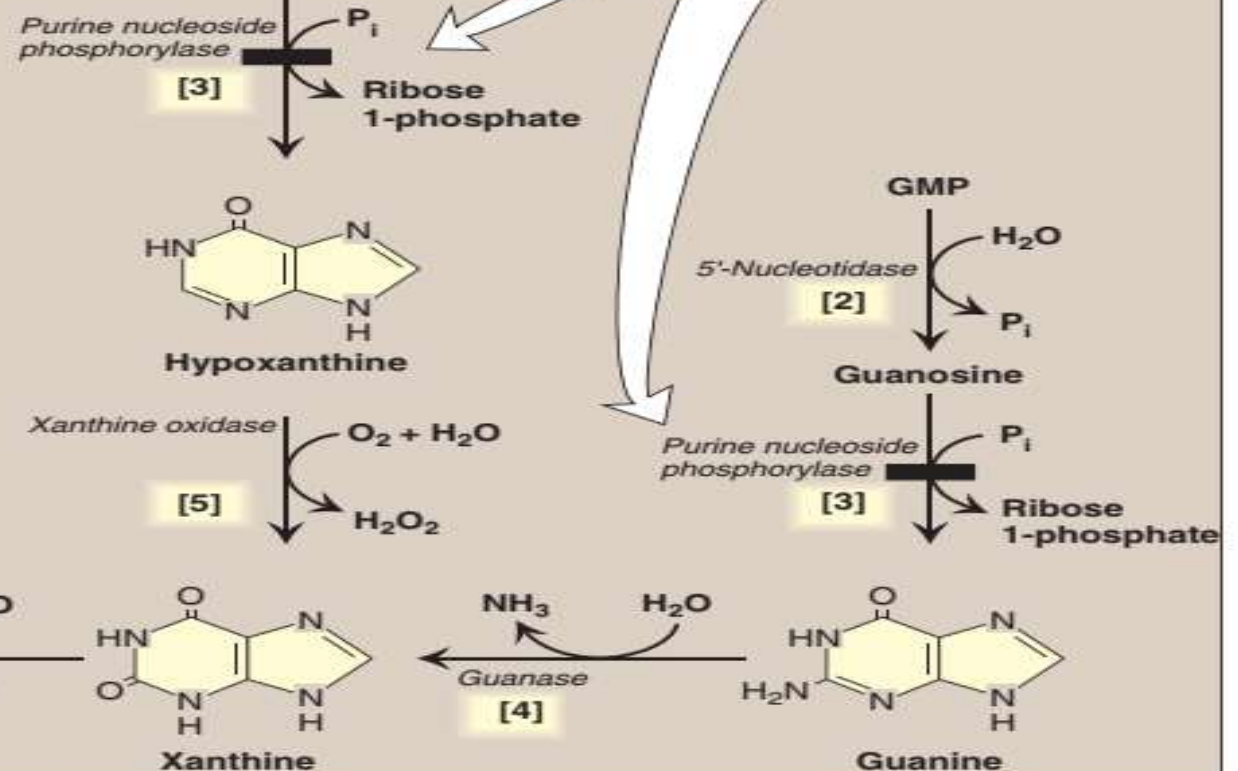
ADENOSINE DEAMINASE (ADA) DEFICIENCY

- This autosomal recessive deficiency causes a type of severe combined immunodeficiency (SCID), involving T-cell, B-cell and NK-cell depletion (lymphocytopenia).
- Untreated ADA-deficient children usually die before 2 years of age from overwhelming infection; treatments include BMT, ERT and gene therapy.



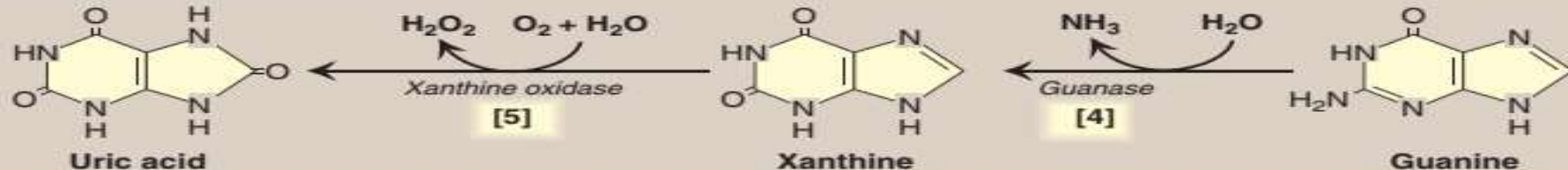
PURINE NUCLEOSIDE PHOSPHORYLASE (PNP) DEFICIENCY

- This autosomal recessive deficiency is rarer and less severe than ADA deficiency.
- Affects only T-cells.
- Characterized by recurrent infections and neuro-developmental delay.



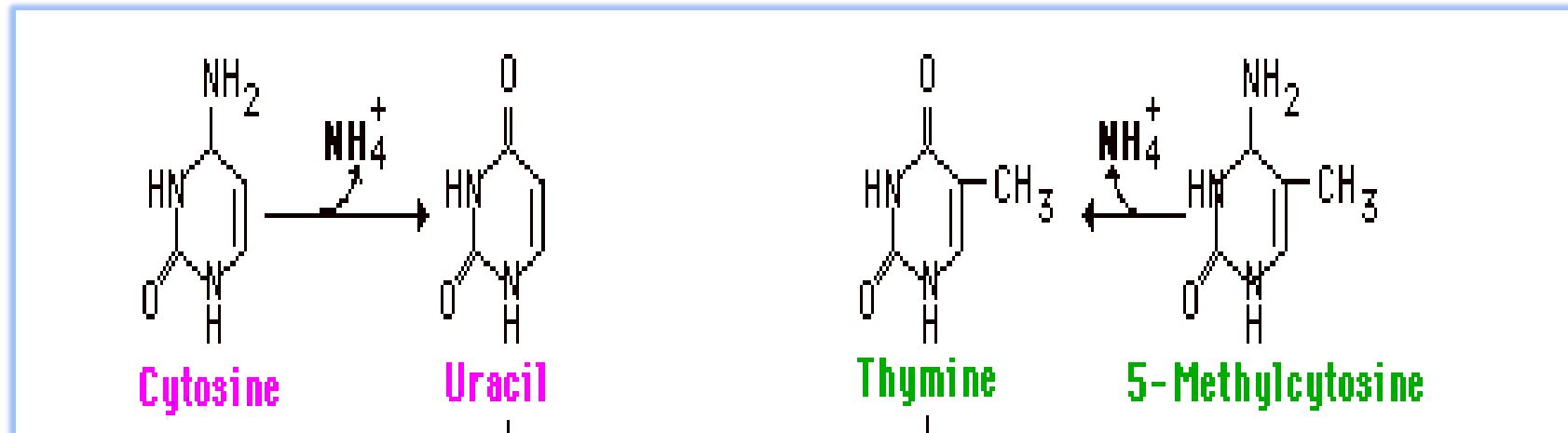
GOUT

- This disorder is characterized by hyperuricemia with recurrent attacks of acute arthritic joint inflammation, caused by deposition of mono-sodium urate crystals.
- In gout, the hyperuricemia results primarily from the underexcretion of uric acid. Overproduction of uric acid is less common, and known causes involve certain inborn errors of metabolism or increased availability of purines.
- Crystal deposition (tophi) may be seen in soft tissue and in kidney (urolithiasis).
- Treatment with allopurinol inhibits *xanthine oxidase*, resulting in an accumulation of hypoxanthine and xanthine—compounds more soluble than uric acid.



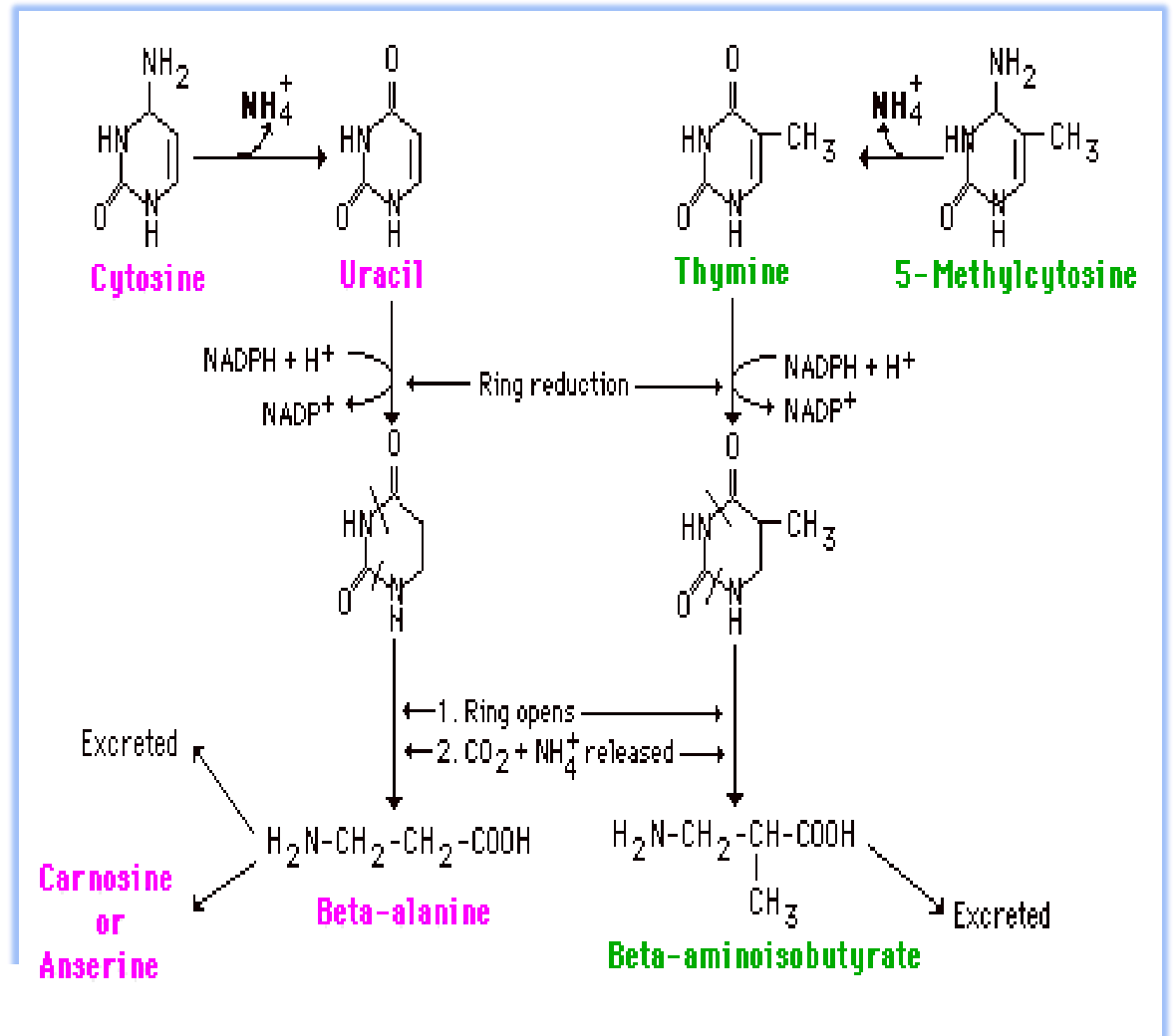
Catabolism of a pyrimidine

- ✓ In contrast to purines, pyrimidines undergo ring cleavage, and the usual end products of catabolism are **β -amino acids** plus **NH_3^+** and **CO_2** .
- ✓ Pyrimidines from nucleic acids or the energy pool are acted upon by **nucleotidases** and **pyrimidine nucleoside phosphorylase** to yield the **free bases**.
- ✓ The 4-amino group of both cytosine and 5-methyl cytosine is **released as ammonia**.
More soluble and useful degradation end product through ring cleavage



Ring Cleavage

- ✓ In order for the **rings (Cytosine and Thymine) to be cleaved**, they must first be **reduced by NADPH**
- ✓ Atoms 2 and 3 of both rings are released as **ammonia and carbon dioxide**
- ✓ The rest of the ring is left as a **beta-amino acid**.
- ✓ **β -amino isobutyrate** from thymine or 5-methyl cytosine is largely excreted
- ✓ **β -alanine** from cytosine or uracil may either be excreted or incorporated into the brain and muscle dipeptides, **carnosine** (his-beta-ala) or **anserine** (methyl his-beta-ala)





- **Carnosine** → endogenous **dipeptide** (**L-Histidine** and **B- alanine**)
- Found in **low millimolar** conc. In **brain** and high conc. In cardiac and **skeletal muscle** (20mM)
- **Deficiency in the brain** associated with **cognitive decline** (in **Alzheimer's**),,, **therapeutic potential** in the **treatment of cognitive disorders**
- **Dietary supplementation** of **carnosine** along with the methylated analog (**anserine**) **improves cognitive function** and **physical activity** among **elderly people**
- They also have a role In skeletal muscle as **buffering** the **lactic acid** due to **anaerobic** respiration

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Thank You!!